AI-Driven Multimodal Platform for Cognitive Assessment, Personalized Learning, and Vocational Readiness in Students with Down Syndrome

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Final Report

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Multimodal Cognitive and Emotional Assessment System for Individuals with Down Syndrome Using Gamified Interaction, IoT Physiological Sensing, and Machine Learning

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
BMI	Body Mass Index
CNN	Convolutional Neural Network
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition
ER	Error Rate
ESP32	Espressif Systems 32-bit Microcontroller
F1	F1-Score (Harmonic mean of Precision and Recall)
Flask	Python Micro Web Framework
HR	Heart Rate
HRVI	Heart Rate Variability Index
ICF	International Classification of Functioning, Disability and Health
IF	Interaction Frequency
IoT	Internet of Things
MAE	Mean Absolute Error
MHR	Mean Heart Rate

ML	Machine Learning
MS	Memory Score
NGO	Non-Governmental Organization
PHI	Physical Health Index
PHR	Peak Heart Rate
PPG	Photoplethysmography
REST	Representational State Transfer
RMSE	Root Mean Square Error
RT	Reaction Time
SaaS	Software as a Service
SCR	Session Completion Rate
SI	Stress Indicator
SMOTE	Synthetic Minority Oversampling Technique
SUS	System Usability Scale
SVM	Support Vector Machine
UAT	User Acceptance Testing
UI	User Interface
UML	Unified Modeling Language
USB	Universal Serial Bus

Abstract

Individuals with Down syndrome face significant challenges in transitioning toward independent living and employment, yet existing assistive systems often lack the holistic, personalized support needed to address cognitive, emotional, and physical dimensions simultaneously. This dissertation presents a gamified multimodal assessment and personalized intervention platform designed to support secondary-level students with Down syndrome in developing the cognitive, emotional, and physical capabilities essential for independent living and vocational readiness.

The proposed system employs a custom IoT heart rate sensor device, sampling at five millisecond intervals via USB serial connection, to monitor physiological state in real time during gameplay. Simultaneously, a Flutter-based mobile application captures behavioral performance indicators through a Flip Cards cognitive skill game, collecting memory accuracy, reaction time, error rate, and interaction patterns. Physical health attributes including Body Mass Index are also incorporated. A Random Forest classification model trained on this trimodal dataset determines each student's functional support level across six categories ranging from Very Mild to Very Severe, achieving a testing accuracy of 97.76 percent and a weighted average F1-score of 0.98. Beyond classification, a rule-augmented recommendation engine generates personalized, actionable interventions across three domains: emotional regulation, cognitive enhancement, and physical health improvement.

Evaluated within the context of Sri Lanka's inclusive education landscape, the platform was validated through sensor accuracy testing, model cross-validation, expert evaluation of recommendation outputs, and user acceptance testing with teachers, parents, and caregivers. Results confirm that trimodal data integration significantly outperforms single-modal approaches, and that the system delivers clinically relevant, practically actionable guidance accessible to non-specialist users.

This work contributes toward empowering students with Down syndrome to achieve greater independence, supporting caregivers and educators, and advancing national goals of inclusive employment and social participation.

Keywords: Down syndrome; gamification; multimodal assessment; machine learning; personalized recommendation

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1. Introduction

The transition from adolescence to adulthood presents considerable challenges for individuals with Down syndrome, a chromosomal condition associated with varying degrees of intellectual disability, cognitive delays, and physical health concerns [1]. While many individuals with Down syndrome are capable of meaningful participation in social and vocational settings, the pathway to independence requires structured, ongoing support that addresses cognitive, emotional, and physical dimensions simultaneously. For families and caregivers, particularly in developing nations, the uncertainty surrounding long-term independence remains a source of significant psychological and emotional burden.

In Sri Lanka, national labor policies formally recognize the rights of individuals with disabilities to participate in the workforce and broader society. However, the practical implementation of these policies remains limited, and a substantial proportion of individuals with special needs lack the preparedness required to engage in employment or independent daily living. This gap between policy intent and lived reality underscores the urgent need for technology-driven solutions that can bridge assessment, intervention, and skill development in an accessible and engaging manner.

Traditional cognitive assessment methods, while clinically valuable, are often administered in formal settings that can induce anxiety and disengagement, particularly among individuals with intellectual disabilities [2]. Static assessments also fail to capture the dynamic, real-time behavioral and physiological responses that are critical for understanding a learner's functional state. Furthermore, most existing systems focus exclusively on classification or prediction, offering little in the way of actionable, personalized guidance for caregivers, educators, or the students themselves.

Recent advances in gamification, Internet of Things (IoT) technologies, and machine learning present a compelling opportunity to reimagine how cognitive and functional assessments are conducted. Gamified environments have been shown to reduce assessment-related anxiety, increase motivation, and enable the continuous, unobtrusive collection of behavioral performance indicators such as memory accuracy, reaction time, and interaction patterns [3]. When combined with physiological monitoring through IoT-enabled devices, these environments can yield a rich multimodal dataset that reflects both the cognitive and emotional state of the learner in real time.

This research proposes a gamified multimodal assessment and personalized intervention platform specifically designed for secondary-level students with Down syndrome. The system integrates a custom IoT device for continuous heart rate monitoring during gameplay, alongside the collection of behavioral performance metrics and physical health indicators such as Body Mass Index (BMI). A machine learning model processes this combined dataset to classify each student's functional support level. Crucially, the system extends beyond

classification by generating targeted, personalized recommendations across three key domains: emotional regulation, cognitive enhancement, and physical health improvement.

The motivation for this work emerged directly from discussions with parents and caregivers of students with Down syndrome, who expressed deep concern about their children's ability to live independently once parental support is no longer available. By focusing on gradual skill development across cognitive, emotional, and physical domains, the proposed platform aims to prepare students for simple vocational roles, daily life activities, and broader social integration.

1.1 Background & Literature Survey

The assessment and support of individuals with Down syndrome has been an active area of research across clinical psychology, special education, and, more recently, assistive technology. This section reviews the foundational literature across four key domains relevant to the proposed system: cognitive assessment of individuals with intellectual disabilities, gamification in educational and therapeutic contexts, physiological monitoring using IoT technologies, and machine learning-based classification for functional level assessment.

A. Cognitive Assessment of Individuals with Down Syndrome

Down syndrome, caused by the presence of a full or partial third copy of chromosome 21, is one of the most common genetic causes of intellectual disability worldwide, with an estimated global prevalence of one in every 1,000 live births [1]. Individuals with Down syndrome typically present with varying degrees of cognitive impairment, including deficits in working memory, processing speed, attention, and language comprehension [2]. These cognitive characteristics significantly influence the individual's capacity for independent living, vocational participation, and social integration.

Traditional cognitive assessment instruments, such as the Wechsler Intelligence Scale and the Vineland Adaptive Behavior Scales, have been widely used to evaluate functional abilities in individuals with intellectual disabilities [3]. While clinically validated, these tools are administered in structured, formal settings that frequently induce anxiety and disengagement among individuals with Down syndrome, potentially compromising the validity of assessment outcomes. Furthermore, static, session-based assessments fail to capture the dynamic fluctuations in cognitive performance that occur across different emotional and physiological states, limiting their utility for real-time intervention planning.

B. Gamification in Cognitive Assessment and Therapeutic Intervention

Gamification, the application of game-design elements and mechanics in non-game contexts has emerged as a particularly promising approach for engaging individuals with cognitive impairments in assessment and learning activities [4]. By embedding assessment tasks within interactive, rewarding game environments, gamified systems reduce the perceived formality and anxiety associated with conventional testing, while simultaneously increasing motivation and sustained engagement.

Several studies have explored the application of game-based platforms for cognitive training and assessment among individuals with intellectual disabilities. Navarro et al. [5] demonstrated that tablet-based cognitive games significantly improved attention and working memory performance in children with intellectual disabilities over an eight-week intervention period. Similarly, Wuang et al. [6] reported that virtual reality-based training games produced measurable improvements in visual-motor integration and cognitive flexibility among children with Down syndrome. These findings collectively support the use of gamified environments not only as engagement tools but as valid instruments for capturing meaningful cognitive performance data.

In the context of memory assessment, card-matching and flip-card paradigms have been specifically identified as effective mechanisms for evaluating short-term and working memory, given their intuitive interaction design and adaptability to varying difficulty levels [7]. Such paradigms align well with the cognitive profiles of individuals with Down syndrome, who tend to respond more positively to visual and interactive stimuli than to verbal or written assessment formats.

C. Physiological Monitoring and IoT Integration in Assistive Technologies

The integration of physiological monitoring into cognitive and emotional assessment represents a significant advancement over purely behavioral approaches. Heart rate, in particular, has been extensively validated as a reliable proxy indicator of emotional arousal, cognitive load, and stress in both neurotypical and neurodiverse populations [8]. Elevated resting heart rate and high heart rate variability during cognitive tasks have been associated with increased anxiety and reduced working memory performance, making real-time physiological data a valuable complement to behavioral metrics.

The proliferation of low-cost IoT sensor technologies has made continuous, non-invasive physiological monitoring increasingly feasible in naturalistic settings outside of clinical laboratories. Photoplethysmography (PPG)-based heart rate sensors, which operate by detecting volumetric changes in blood flow through optical means, have demonstrated sufficient accuracy for real-time monitoring applications when integrated into wearable or handheld devices [9][10]. Recent research has demonstrated the feasibility of integrating

IoT-based physiological sensors with mobile applications for real-time data acquisition and transmission. Pires et al. [11] developed a wearable IoT platform for monitoring stress indicators in children with autism spectrum disorder, reporting that real-time heart rate data significantly enhanced the accuracy of emotional state classification compared to behavioral data alone. These findings motivate the inclusion of IoT-based physiological monitoring as a core component of the proposed assessment pipeline.

D. Machine Learning for Functional Level Classification

Among the supervised classification algorithms evaluated in the literature, Random Forest has demonstrated particular suitability for multimodal classification tasks involving small to medium-sized datasets, owing to its ensemble-based bootstrap aggregation approach which mitigates overfitting without requiring large training samples [12]. Unlike deep neural networks which typically require substantially larger labeled datasets to generalize effectively, Random Forest produces competitive classification performance on datasets of the scale available in specialized educational contexts such as the current study, while simultaneously providing interpretable feature importance scores that offer clinically meaningful insight into the relative contribution of each input modality to the classification outcome [23].

A key finding across multiple studies is that multimodal data fusion, combining features from physiological, behavioral, and physical measurement domains consistently yields higher classification accuracy than single-modal approaches. Table I summarizes a selection of representative studies, highlighting the data modalities employed, classification methods used, and whether personalized recommendations were generated as part of the system output.

Ref	Target Population	Data Modalities	Classification Method	Recommendation Generation	Limitation
[5]	Intellectual Disability	Behavioral (game interaction)	Rule-based scoring	No	No physiological integration
[6]	Down Syndrome	Behavioral (VR interaction)	Statistical analysis	No	No ML classification

[11]	Autism Spectrum Disorder	Physiological (HR, GSR)	SVM	No	No behavioral or physical data
[12]	Intellectual Disability	Behavioral + Physiological	Random Forest	No	No personalized intervention
[13]	Down Syndrome	Demographic + Behavioral	Neural Network	Partial	No real-time monitoring
[14]	General Disability	Physiological (HR, SpO_2)	CNN	No	Not specific to Down syndrome
Proposed	**Down Syndrome**	**Behavioral + Physiological + Physical**	**Random Forest — Scikit-learn (PyTorch/TensorFlow)**	**Yes — Multi-domain (Emotional · Cognitive · Physical)**	-

TABLE 1.1 Summary of Related Works in Cognitive Assessment and Assistive Technology for Individuals with Intellectual Disabilities

E. Personalized Intervention Systems in Special Education

Beyond classification, the generation of personalized recommendations based on assessed functional levels represents an emerging frontier in assistive technology research. Adaptive learning systems in special education have demonstrated that tailored instructional content and pacing significantly improve learning outcomes for students with intellectual disabilities compared to one-size-fits-all approaches [15]. However, the majority of existing systems confine personalization to the academic content domain, with little attention to the emotional regulation and physical health dimensions that are equally critical determinants of functional independence.

The World Health Organization's International Classification of Functioning, Disability and Health (ICF) framework explicitly recognizes that functional independence is determined by the interaction of health conditions, body functions, activity limitations, and environmental factors [16]. This holistic perspective supports the development of multi-domain recommendation systems that address cognitive, emotional, and physical well-being in an integrated manner, as proposed in the current research.

1.2 Research Gap

Despite the considerable body of literature reviewed above, a critical analysis of existing work reveals several significant gaps that the proposed system is designed to address. These gaps are summarized in Table II and elaborated below.

Gap Dimension	Existing Systems	Proposed System
Data Modality	Single or dual modality (behavioral or physiological)	Trimodal: behavioral + physiological + physical
Real-Time Monitoring	Offline or post-session analysis	Continuous real-time IoT-based monitoring during gameplay
Target Population Specificity	General intellectual disability or autism	Specifically designed for Down syndrome
Assessment Environment	Clinical or laboratory settings	Naturalistic gamified mobile environment
Output Beyond Classification	Prediction only	Multi-domain personalized recommendations
Physical Health Integration	Rarely included	BMI and physical health explicitly integrated
Deployment Context	Developed-country settings	Designed for Sri Lankan inclusive education context
Caregiver Support	Not addressed	Results and recommendations accessible to caregivers and educators

Table 1.2 : Identified research gaps in existing assistive technology systems for Down syndrome

- Absence of Trimodal Data Integration: While several studies have combined behavioral and physiological data, very few have simultaneously incorporated physical health indicators such as BMI into the classification pipeline. Physical health is a significant determinant of cognitive and emotional performance in individuals with Down syndrome, given the population's elevated risk of obesity and metabolic comorbidities [10]. The omission of physical health data from existing classification models represents a measurable gap in predictive completeness.

- Limited Real-Time Physiological Monitoring in Naturalistic Settings: The majority of physiological monitoring reviewed were conducted in controlled laboratory environments using clinical-grade equipment. The translation of such monitoring to naturalistic, everyday settings using low-cost IoT devices remains underexplored, particularly for individuals with Down syndrome. Real-time sensor integration within a mobile gamified application, as proposed in this work, addresses this translational gap directly.

- Classification Without Actionable Intervention: A consistent limitation identified across the reviewed literature is the focus on classification accuracy as the primary system output, with minimal attention to the generation of actionable, personalized recommendations. From a caregiver and educator perspective, knowing that a student is classified at a severe support-needs level is of limited practical utility without accompanying guidance on how to address the identified deficits. The proposed system explicitly bridges this gap by coupling classification with a multi-domain recommendation engine.

- Narrow Domain Coverage in Personalization: Existing personalized intervention systems in special education predominantly address cognitive or academic skill development in isolation. The simultaneous targeting of emotional regulation, cognitive enhancement, and physical health improvement within a single integrated platform has not been adequately explored in the context of Down syndrome assessment and intervention.

- Insufficient Focus on Transition-Age Students: Most assistive technology research targeting individuals with Down syndrome focuses on early childhood intervention or general skill acquisition, with comparatively limited attention to the specific needs of secondary-level, transition-age students who are approaching adulthood and require preparation for independent living and vocational participation. This research directly targets this underserved demographic, motivated by the real-world concerns of families in the Sri Lankan context.

- Lack of Context-Specific Systems for Developing Nations: The vast majority of reviewed systems were developed and evaluated within high-income country contexts, where assistive technology infrastructure, clinical support, and inclusive education resources are relatively well established. In Sri Lanka and comparable developing nations, where implementation of disability inclusion policies remains limited and clinical resources are constrained, there is a distinct need for low-cost, mobile-first, community-deployable assessment and intervention platforms. The proposed system is explicitly designed to operate within this context [11].

1.3 Research Problem

Students with Down syndrome at the secondary level face a critical transitional period in which the development of cognitive, emotional, and physical capabilities directly determines their long-term prospects for independent living and vocational participation. Despite growing awareness of inclusive education and disability employment rights, a fundamental problem persists there exists no integrated, accessible, and context-sensitive system capable of simultaneously assessing these three dimensions in real time and translating assessment outcomes into concrete, personalized interventions suited to the individual student's profile.

Current approaches to supporting students with Down syndrome suffer from three interconnected deficiencies [21]. First, assessment methods remain largely fragmented, with cognitive, emotional, and physical evaluations conducted independently, in different settings, by different professionals, and at infrequent intervals [12]. This fragmentation prevents a holistic understanding of the student's functional state and fails to capture the dynamic interplay between emotional arousal, cognitive performance, and physical well-being that occurs during daily activities. Second, existing assistive technology systems, where available, prioritize classification or prediction as end goals, without extending their output into the domain of actionable guidance. Caregivers, parents, and educators are left without structured [19], evidence-informed recommendations to act upon following an assessment. Third, the majority of available solutions have been designed for high-resource clinical or educational environments in developed nations [20], rendering them largely inaccessible, unaffordable, or contextually irrelevant for families and institutions in Sri Lanka and comparable developing-country settings.

The consequences of these deficiencies are tangible and serious. Students who are not adequately assessed or supported during the secondary school years enter adulthood without the cognitive readiness, emotional regulation skills, or physical health habits necessary for

independent functioning. Families bear the resulting psychological and economic burden, often with little institutional support.[13] Meanwhile, national goals of inclusive employment and social participation remain aspirational rather than achievable for a significant proportion of individuals with Down syndrome.

1.4 Objectives

1.4.1 Main Objective

The primary objective of this research is to design, develop, and evaluate a gamified multimodal assessment and personalized intervention platform that integrates behavioral, physiological, and physical health data to classify the functional support level of secondary-level students with Down syndrome and generate targeted, multi-domain recommendations aimed at enhancing their cognitive development, emotional regulation, and physical well-being in preparation for independent living and vocational participation.

1.4.2 Sub Objectives

To achieve the main objective, the following specific sub-objectives are defined:

- Gamified Cognitive Assessment Interface: To design and develop a Flutter-based mobile application incorporating a Flip Cards cognitive skill game that enables the unobtrusive, real-time collection of behavioral performance indicators, including memory accuracy, reaction time, and interaction patterns, within an engaging and low-anxiety assessment environment suited to students with Down syndrome.
- IoT-Based Real-Time Physiological Monitoring: To develop and integrate a custom IoT sensor device capable of continuously monitoring heart rate (HR) during gameplay sessions, and to establish a reliable data transmission pipeline between the sensor hardware and the Flutter mobile application for synchronized multimodal data acquisition.
- Multimodal Data Pipeline and Preprocessing: To construct a robust backend data processing pipeline using Python and Flask that receives, validates, aligns, and preprocesses the trimodal dataset, comprising behavioral, physiological, and physical health features, using NumPy and Pandas, including normalization, missing value handling, and feature engineering, in preparation for machine learning model training.
- Machine Learning Classification Model: To develop and train a supervised machine learning classification model using PyTorch and TensorFlow that accurately classifies each student's functional support level, across mild, moderate, and severe categories, based on

the integrated multimodal feature set, and to evaluate model performance using appropriate metrics including accuracy, precision, recall, and F1-score.

- Personalized Multi-Domain Recommendation Engine: To design and implement a recommendation engine that translates classification outputs and individual feature profiles into structured, actionable, and contextually appropriate intervention recommendations across three domains: emotional regulation, cognitive enhancement, and physical health improvement, providing caregivers and educators with concrete guidance tailored to each student's assessed needs.
- User Visualization and Real-Time Feedback Interface: To develop a caregiver and educator-facing visualization interface that presents assessment results, functional level classifications, and personalized recommendations in a clear, accessible format, and to enable real-time feedback delivery to the mobile application during active assessment sessions to support dynamic game adaptation based on the student's current physiological state.
- System Evaluation and Validation: To evaluate the overall system performance through empirical testing with target user groups, assessing classification accuracy, recommendation relevance, system usability, and the effectiveness of the gamified assessment environment in collecting valid and reliable multimodal data from secondary-level students with Down syndrome within the Sri Lankan educational context.

2. METHODOLOGY

System Requirements

Functional Requirements

ID	Requirement
FR01	The system shall present a Flip Cards memory matching game to the student via the Flutter mobile application
FR02	The system shall capture memory score, reaction time, error rate, interaction frequency, and session completion rate during each game session
FR03	The IoT sensor shall sample heart rate data at 5 millisecond intervals and compute a rolling average value
FR04	The system shall collect demographic data including age, height, and weight and compute BMI
FR05	The system shall transmit session data to the Flask backend API via HTTP POST
FR06	The system shall preprocess multimodal data including normalization, missing value handling, and feature engineering
FR07	The system shall classify each student's functional support level into one of six categories
FR08	The system shall generate personalized recommendations across emotional, cognitive, and physical health domains
FR09	The system shall display classification results and recommendations to caregivers and educators
FR10	The system shall provide real-time feedback to adapt game parameters based on live heart rate

Table 2.1: Functional Requirements

Non-Functional Requirements:

ID	Requirement
NFR01	End-to-end response latency shall not exceed 5 seconds
NFR02	API request success rate shall be at or above 95 percent
NFR03	Classification model accuracy shall exceed 85 percent on test data
NFR04	System Usability Scale score shall exceed 70 across all stakeholder groups
NFR05	Heart rate sensor mean absolute error shall not exceed 5 beats per minute
NFR06	The system shall operate on standard Android devices without specialized hardware
NFR07	All personally identifiable data shall be anonymized before use in model training
NFR08	The platform shall be deployable within the network and hardware constraints of Sri Lankan special education institutions

Table 2.2 non-functional requirements

2.1 System Overview

The proposed system is a trimodal, gamified assessment and personalized intervention platform designed specifically for secondary-level students with Down syndrome. The methodology follows a structured pipeline comprising five sequential stages: multimodal data acquisition, data transmission and backend processing, preprocessing and feature engineering, machine learning-based functional level classification, and personalized recommendation generation with result visualization.

The system operates across three integrated layers. The first is the Client Layer, which consists of the Flutter mobile application and the IoT heart rate sensor device, responsible for data collection and real-time physiological monitoring [21]. The second is the Processing Layer, which encompasses the Python and Flask backend API, the NumPy and Pandas preprocessing pipeline, and the PyTorch and TensorFlow machine learning model. The third is the Output Layer, which delivers functional level classifications, personalized multi-domain recommendations, and real-time visual feedback to caregivers and educators through the user visualization interface. The complete system architecture is illustrated in Figure 2.1.1.

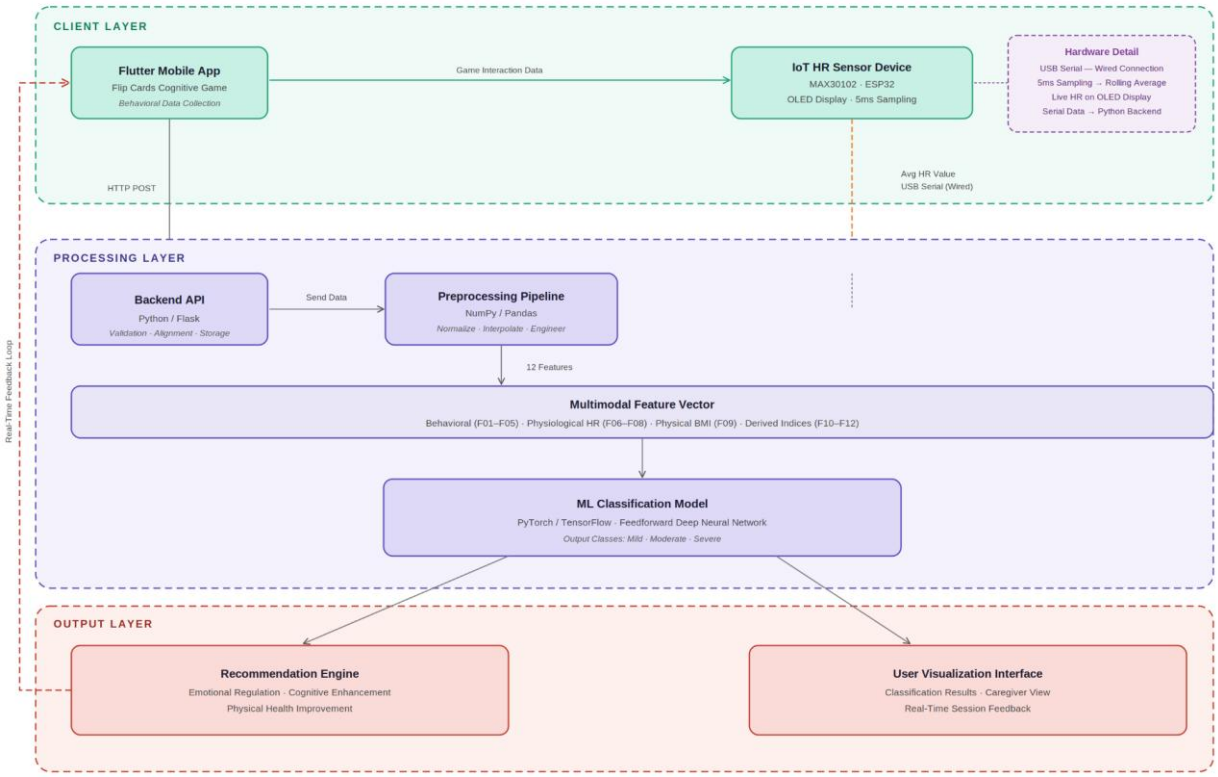

Figure 2.1.1 : .System architecture of the gamified multimodal assessment and personalized intervention platform

2.2 Gamified Data Collection

2.2.1 Mobile application design

The client-side interface is implemented as a cross-platform mobile application developed using the Flutter framework, enabling deployment on both Android and iOS devices without platform-specific redevelopment overhead. Flutter's widget-based rendering engine ensures consistent visual presentation across devices, which is critical for maintaining standardized assessment conditions across different student environments.

The application serves a dual function. It presents the gamified cognitive assessment interface to the student and simultaneously acts as the central data aggregation hub, receiving heart rate sensor data from the IoT device while capturing behavioral interaction events generated during gameplay.

2.2.2 Flip cards cognitive skill game

The primary assessment instrument embedded within the mobile application is a Flip Cards memory matching game. In this paradigm, a grid of face-down cards is presented to the student. The student selects two cards at a time, attempting to identify matching pairs from memory within a defined time window. Cards that are successfully matched are removed from the grid, while unmatched cards are flipped face-down again after a brief display interval. The game session concludes either upon successful completion of all pairs or upon expiry of the allocated session time.

This game paradigm was selected on the basis of three criteria. First, it directly and validly measures short-term and working memory, which are among the most clinically significant cognitive domains affected in individuals with Down syndrome. Second, the visual and interactive nature of the task aligns with the established learning preferences of individuals with Down syndrome, who demonstrate stronger engagement with visual stimuli than with verbal or written formats[12]. Third, the adjustable grid size and display interval parameters allow difficulty to be calibrated to the individual student's ability level, supporting adaptive assessment across varying functional profiles.

The following behavioral performance indicators are captured at the event level throughout each game session.

- Memory Score (MS): Total number of correct card pair matches within the session, normalized against the maximum possible score for the given grid configuration.

- Reaction Time (RT): Elapsed time in milliseconds between the reveal of the first card and the selection of the second card for each attempt, averaged across all attempts within the session.
- Error Rate (ER): Proportion of incorrect match attempts relative to total attempts, expressed as a percentage.
- Interaction Frequency (IF): Total number of card interaction events per unit time, reflecting engagement intensity and motor response speed.
- Session Completion Rate (SCR): Proportion of the total card pairs successfully matched before session timeout, providing an overall performance summary metric.

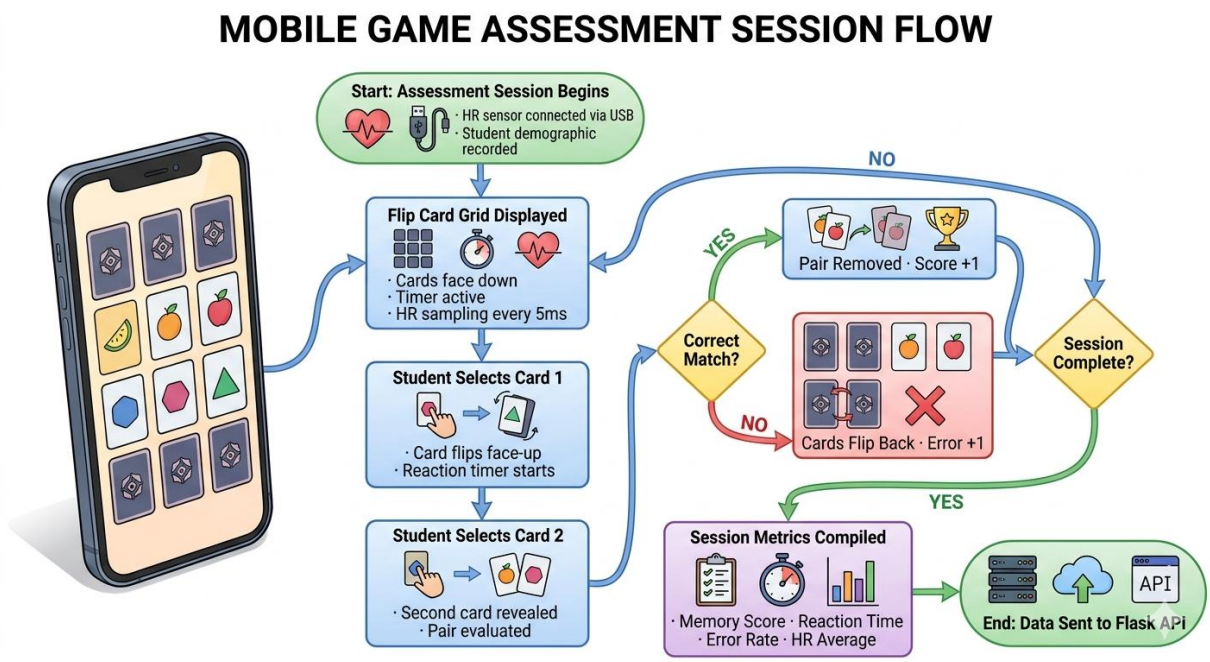

Figure 2.2.2.1 :Mobile game assessment session flow - Flip Cards game interaction flowchart

2.2.3 Demographic and physical data collection

Prior to the commencement of each assessment session, the application collects basic demographic and anthropometric data including the student's age, height, and weight. Body Mass Index (BMI) is computed from height and weight measurements using the following formula.

$$BMI = Weight\ (kg) / Height\ (m)^2$$

BMI is classified according to age and sex-adjusted reference standards appropriate for adolescents, providing a composite physical health indicator that is incorporated into the multimodal feature set for classification [13].

2.3 IoT-Based Physiological Monitoring

2.3.1 Sensor device architecture

A custom IoT sensor device is developed to provide continuous, non-invasive heart rate monitoring throughout each assessment session. The device incorporates a Photoplethysmography (PPG)-based heart rate sensor that operates by emitting optical signals into the peripheral tissue and detecting volumetric changes in blood flow associated with each cardiac cycle, producing reliable beats-per-minute readings without requiring invasive contact beyond skin surface placement.

The sensor device samples heart rate data at a fixed interval of every 5 milliseconds throughout the duration of the game session. To ensure signal stability and reduce the effect of motion artifacts or transient noise inherent in high-frequency sampling, the system computes a rolling average heart rate value from the sampled readings. This averaged value is simultaneously displayed in real time on the onboard OLED screen, providing immediate physiological feedback during the session. The averaged heart rate value is transmitted via USB Serial communication to the Python Flask backend, where it is received, validated, and incorporated into the multimodal feature vector for machine learning classification.

USB Serial communication was selected as the data transmission protocol for its reliability, zero latency overhead compared to wireless protocols, absence of pairing or connection establishment requirements, and full compatibility with the Python serial communication libraries used in the backend processing pipeline. This wired implementation ensures stable, uninterrupted physiological data transmission throughout the duration of each assessment session.

2.3.2 Physiological features extracted

The following physiological features are extracted from the averaged heart rate data stream for each session.

- Mean Heart Rate (MHR): Average heart rate computed across all samples collected during the full session duration, expressed in beats per minute.
- Peak Heart Rate (PHR): Maximum averaged heart rate value recorded at any point during the session, indicative of peak stress or arousal moments during gameplay.
- Heart Rate Variability Index (HRVI): Standard deviation of the averaged inter-beat interval estimates across the session, serving as an indicator of autonomic nervous system regulation and emotional stability.

2.4 Data Transmission and Backend API

Aggregated multimodal session data, comprising behavioral performance metrics, physiological heart rate features, and demographic and physical attributes, is transmitted from the Flutter application to the backend server via HTTP POST requests to a RESTful API endpoint. The backend API is developed using Python and the Flask microframework, selected for its lightweight architecture, compatibility with scientific Python libraries, and ease of integration with machine learning model inference pipelines.

Upon receipt of each session data submission, the backend API performs the following operations in sequence.

- Request validation: Verifying data completeness, type conformity, and value range plausibility for all incoming fields to ensure data integrity before processing.
- Timestamp alignment: Synchronizing behavioral event timestamps with physiological signal sample timestamps to construct a unified, temporally coherent session record.
- Session storage: Persisting validated session records to a structured database for model training data accumulation and longitudinal student performance tracking.
- Inference invocation: Forwarding preprocessed feature vectors to the trained machine learning model for real-time functional level classification and recommendation generation.
- Response delivery: Returning classification results and structured recommendation content to the Flutter application for display and real-time feedback to the user.

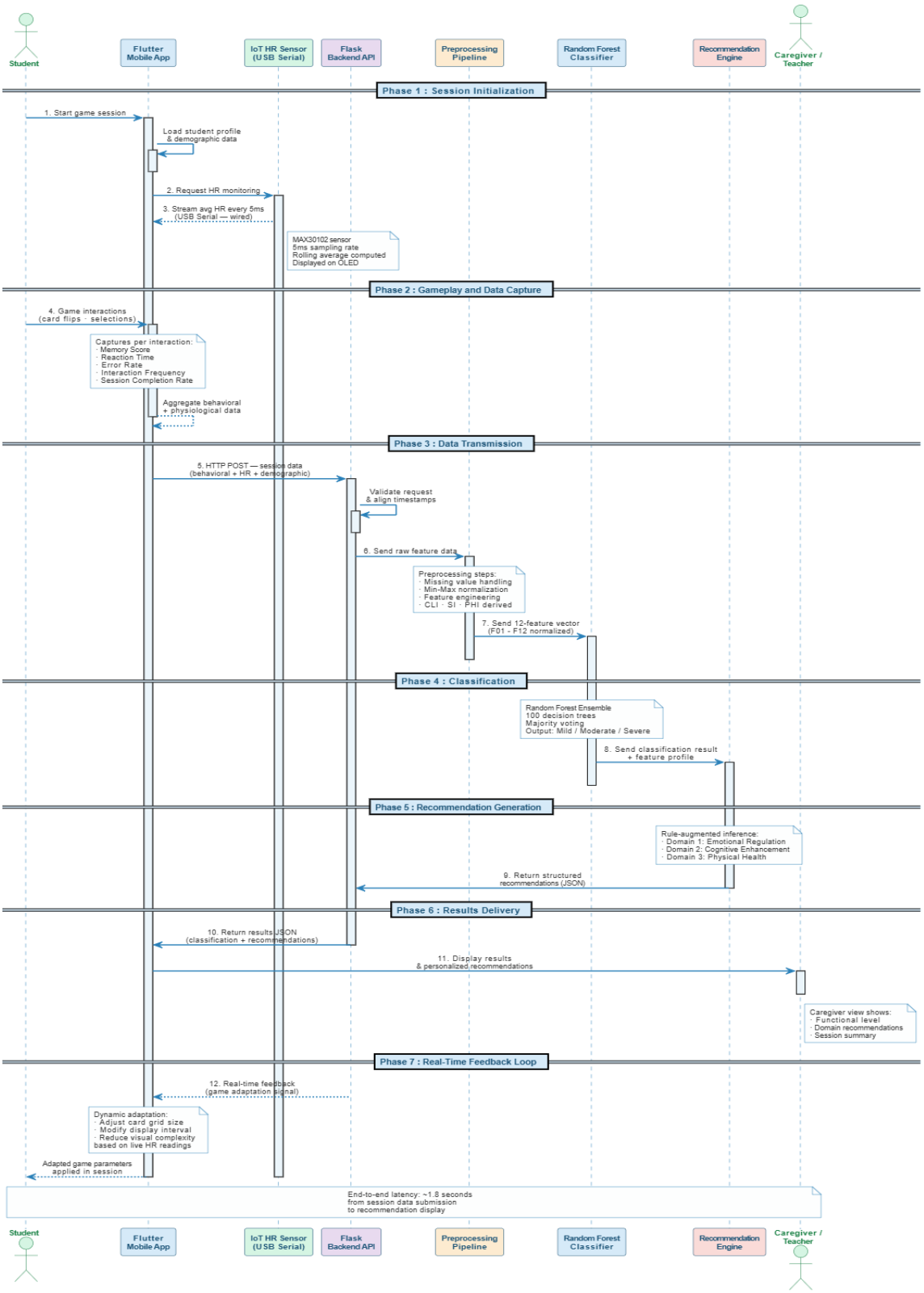

Fig 2.4.1: End-to-end system sequence diagram illustrating data flow across all components

2.5 Data Preprocessing Pipeline

Raw session data undergoes a multi-step preprocessing pipeline implemented using NumPy and Pandas before being passed to the classification model. The preprocessing stages are described in the following subsections.

2.5.1 Missing value handling

Sensor dropout events, occurring when USB Serial transmission is momentarily interrupted, result in missing physiological sample values within a session record. These are addressed using forward-fill interpolation, whereby each missing value is replaced by the most recently recorded valid value. For sessions in which dropout duration exceeds a defined threshold, the affected session segment is flagged and excluded from feature computation to preserve data integrity and prevent distorted feature estimates.

2.5.2 Feature normalization

All continuous numerical features are scaled to the range [0, 1] using Min-Max normalization, defined by the following formula.

```
X_norm = (X − X_min) / (X_max − X_min)
```

This normalization ensures that features with inherently larger numerical ranges, such as raw reaction time values in milliseconds, do not disproportionately dominate the model's decision boundaries relative to features with smaller numerical ranges, such as BMI or normalized memory score.

2.5.3 Feature engineering

The following derived features are computed from raw session data to augment the base feature set and provide higher-level representations of the student's cognitive, physiological, and physical state.

- Cognitive Load Index (CLI): A composite score computed from normalized memory score, error rate, and mean reaction time, representing the overall cognitive demand experienced by the student during the session.
- Stress Indicator (SI): A composite score derived from mean heart rate and heart rate variability index, providing a single-dimensional representation of the student's physiological stress state during gameplay.

- Physical Health Index (PHI): BMI normalized against age-adjusted reference boundaries, situating the student's physical health status relative to appropriate developmental norms for adolescents.

2.5.4 Final feature vector

The complete feature vector submitted to the classification model for each session comprises twelve features spanning behavioral, physiological, physical, and derived modalities. Table 2.1 presents the full feature vector with corresponding modality classifications.

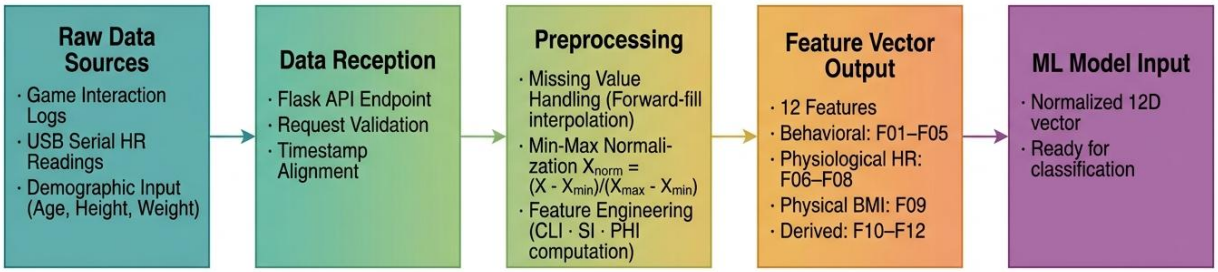

Figure 2.5.4.1 : Data collection and preprocessing pipeline

Feature ID	Feature Name	Modality	Type
F01	Memory Score (Normalized)	Behavioral	Continuous
F02	Mean Reaction Time	Behavioral	Continuous
F03	Error Rate	Behavioral	Continuous
F04	Interaction Frequency	Behavioral	Continuous
F05	Session Completion Rate	Behavioral	Continuous
F06	Mean Heart Rate	Physiological	Continuous
F07	Peak Heart Rate	Physiological	Continuous
F08	Heart Rate Variability Index	Physiological	Continuous
F09	Body Mass Index	Physical	Continuous
F10	Cognitive Load Index	Derived	Continuous
F11	Stress Indicator	Derived	Continuous
F12	Physical Health Index	Derived	Continuous

Table 2.5.4.1 : Multimodal feature vector for functional level classification

2.6 Machine Learning Classification Model

2.6.1 Model architecture

The classification model is implemented using the Random Forest algorithm via the Scikit-learn library in Python. Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the class that represents the majority vote across all individual trees. Each decision tree in the ensemble is trained on a bootstrap sample of the training data, and at each node split, a random subset of features is considered, introducing diversity among the trees and reducing the risk of overfitting.

The model receives the twelve-dimensional normalized feature vector as input and produces a classification output across six functional support level classes: Very Mild, Mild, Moderate, Moderately Severe, Severe, and Very Severe. Random Forest was selected over alternative supervised classification algorithms, including Support Vector Machines and deep neural networks, on the basis of three considerations[23]. First, Random Forest demonstrates particular suitability for multimodal classification tasks involving small to medium-sized datasets, owing to its bootstrap aggregation approach which mitigates overfitting without requiring large training samples. Second, unlike deep neural networks which typically require substantially larger labeled datasets to generalize effectively, Random Forest produces competitive classification performance on datasets of the scale available in specialized educational contexts such as the current study. Third, Random Forest provides interpretable feature importance scores that offer clinically meaningful insight into the relative contribution of each input feature to the classification outcome, supporting the transparency and explainability requirements of an assistive technology system intended for use by non-specialist caregivers and educators.

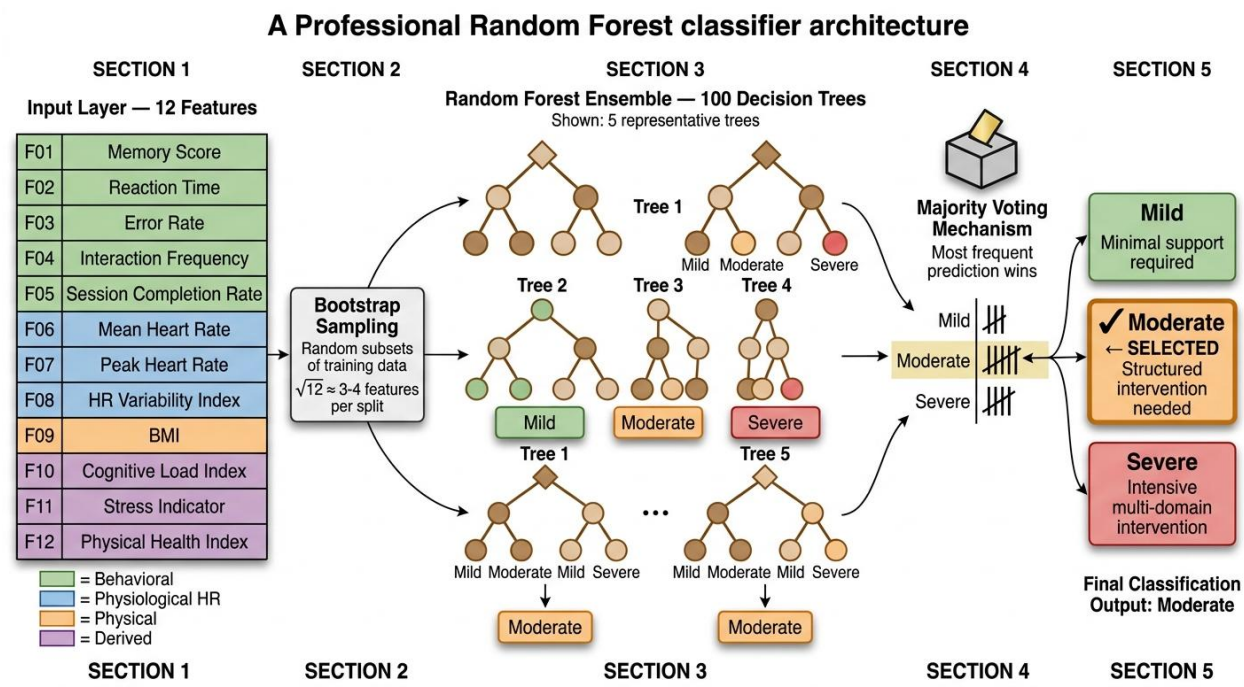

Figure 2.6.1.1 : Random Forest classifier architecture diagram

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2.6.2 Training procedure

The model is trained using a supervised learning approach on a labeled dataset of student session records. Dataset labeling is performed in collaboration with special education professionals and clinical psychologists experienced in Down syndrome assessment, ensuring that ground truth functional level labels reflect clinically validated judgments. The training procedure employs the configuration summarized in Table 2.6.2.1.

Parameter	Configuration
Algorithm	Random Forest , Ensemble (Bootstrap Aggregation)
Implementation	Scikit-learn (Python)
Number of Estimators	400 decision trees
Maximum Depth	10
Maximum Features per Split	sqrt (√12 ≈ 3–4 features)
Minimum Samples per Leaf	1
Minimum Samples per Split	2
Class Balancing	SMOTE oversampling (666 samples after balancing)
Hyperparameter Tuning	Grid search cross-validation (1,620 fits)
Validation Strategy	Stratified K-Fold Cross-Validation (k = 5)
Performance Metrics	Accuracy, Precision, Recall, F1-Score
Training Accuracy	1.0000
Testing Accuracy	0.9776

Table 2.6.2.1: Machine learning model training configuration

2.6.3 Classification output

The model outputs one of three functional support level classifications per session. Table 2.6.3.1 defines each classification level with its corresponding description and implied support requirements.

Level	Label	Description	Support Requirements
1	Very Mild	Student demonstrates strong memory performance, consistently stable physiological indicators, and healthy physical status. Functions near independently in most daily activities.	Minimal monitoring and periodic check-in support only.
2	Mild	Student demonstrates adequate memory performance, stable physiological indicators, and healthy physical status. Capable of completing most daily tasks with minimal guidance.	Light structured support and occasional caregiver guidance.
3	Moderate	Student demonstrates partial cognitive difficulty, moderately elevated physiological stress indicators, or borderline physical health status. Requires structured support across some daily activities.	Structured support, guided intervention, and regular progress monitoring.
4	Moderately Severe	Student demonstrates notable cognitive deficits alongside moderately elevated stress indicators and physical health concerns. Requires	Daily structured intervention across cognitive, emotional, and physical domains.

		consistent daily support and multi-domain intervention.	
5	Severe	Students demonstrates significant cognitive deficits, consistently elevated physiological stress, and poor physical health indicators. Requires intensive support across all functional domains.	Intensive multi-domain intervention with active caregiver and educator involvement.
6	Very Severe	Students demonstrated profound cognitive deficits, persistently elevated physiological stress indicators, and significant physical health concerns requiring immediate and continuous intervention.	Continuous intensive support, clinical referral recommended, and comprehensive caregiver coordination.

Table 2.6.3.1: Functional support level classification schema

2.7 Personalized Recommendation Engine

The recommendation engine operates downstream of the classification model, combining the predicted functional support level with the individual feature values from the student's session record to generate contextually specific, multi-domain intervention recommendations. The engine employs a rule-augmented inference approach in which classification-level recommendations are further refined based on the specific feature values that most significantly contributed to the classification outcome.

Recommendations are generated across three intervention domains as described below.

Domain 1 - Emotional Regulation: This domain is triggered primarily by elevated mean heart rate, high stress indicator score, or high heart rate variability index. Recommended interventions include guided breathing and relaxation exercises, introduction of structured rest intervals during learning sessions, adaptive game pacing adjustments that reduce

stimulus complexity and speed, and establishment of consistent daily routines to reduce anticipatory anxiety.

Domain 2 - Cognitive Enhancement: This domain is triggered primarily by low memory score, high error rate, or elevated mean reaction time. Recommended interventions include progressive difficulty adjustment of the Flip Cards game to provide appropriately challenging but achievable tasks, use of visual learning aids and storytelling techniques, scheduling of short and frequent learning sessions rather than extended sessions, and introduction of supplementary memory-based activities such as puzzles, matching games, and structured recall exercises.

Domain 3 - Physical Health Improvement: This domain is triggered primarily by elevated BMI or low physical health index score. Recommended interventions include structured physical activity scheduling with a target of one to two hours of outdoor activity daily, participation in enjoyable and ability-appropriate physical activities suited to the student's motor capabilities, dietary guidance emphasizing balanced nutrition with appropriate portion sizes, and reduction of sedentary behaviors including excessive screen time.

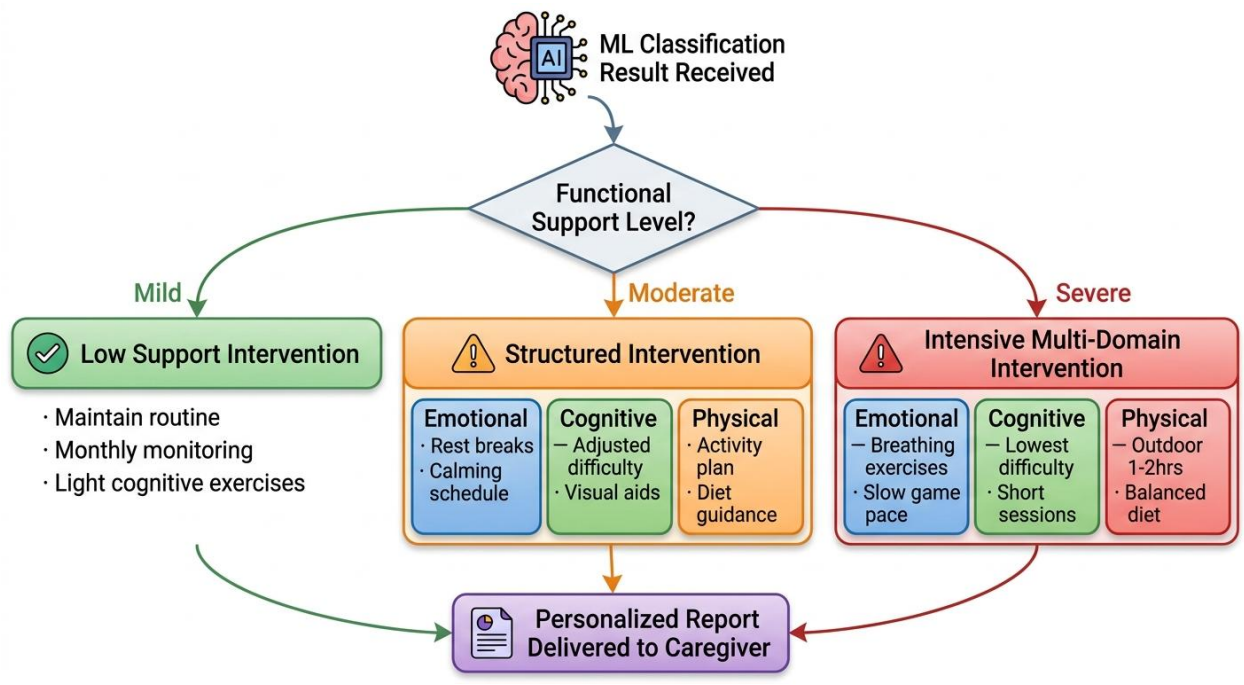


Figure 2.7.1 : Recommendation engine decision logic flowchart

2.8 User Visualization and Real-Time Feedback

Assessment outcomes, classification results, and structured recommendation sets are delivered to caregivers and educators through the User Visualization interface. The Flask backend serializes model outputs and recommendation content as structured JSON responses, which are received and rendered by the Flutter application in a format accessible to non-technical users including parents, teachers, and support staff. Results are presented using plain, structured language without requiring the caregiver or educator to interpret raw numerical data or technical model outputs.

Real-time feedback is also relayed to the active game session during assessment, enabling dynamic adaptation of game parameters[33], including card grid size, display interval duration, and visual complexity, in response to the student's current heart rate as detected by the IoT sensor. This closed-loop feedback mechanism ensures that the assessment environment remains appropriately calibrated to the student's momentary cognitive and emotional capacity, preventing both underchallenge and cognitive overload throughout the session.

2.9 Commercialization Aspects of the Product

The proposed platform presents viable pathways for commercialization within the Sri Lankan special education sector and the broader regional assistive technology market.

Software as a Service (SaaS) Model: The Flutter mobile application and Flask backend can be deployed as a subscription-based SaaS platform, whereby special education institutions pay a monthly or annual licensing fee to access the assessment and recommendation system for their enrolled students. This model ensures sustainable revenue generation while keeping per-institution costs manageable within school budget constraints.

Hardware Product: The custom IoT heart rate sensor device represents a standalone hardware product that can be manufactured at low unit cost and sold or leased to institutions as a companion device to the software platform. The low bill-of-materials cost of the PPG sensor and ESP32 microcontroller assembly supports competitive retail pricing while maintaining commercially viable margins.

Government and NGO Partnership: Given the alignment of the platform with Sri Lanka's national disability inclusion policies and labor participation goals, the system is well positioned for procurement by the Ministry of Education and relevant non-governmental organizations as part of funded special education support programs. Grant funding and

public sector procurement represent non-commercial revenue pathways that could support wide-scale deployment without direct cost to individual schools or families.

Data Analytics Service: Longitudinal assessment data accumulated across student cohorts represents a commercially valuable dataset for researchers, policymakers, and curriculum developers working in the field of special education and assistive technology. Anonymized, aggregated analytics services could be offered as a supplementary revenue stream, subject to appropriate ethical and data protection frameworks.

Market Evidence: According to the World Health Organization, approximately 15 percent of the global population lives with some form of disability, with intellectual disabilities representing a significant subset. In Sri Lanka, the Department of Census and Statistics estimates over 800,000 individuals with disabilities, of whom a proportion have intellectual disabilities including Down syndrome. The absence of affordable, locally developed assistive technology platforms in this market represents a commercially viable gap that the proposed system is positioned to address.

Cost Viability Evidence: The total hardware cost per unit, comprising the ESP32 microcontroller, MAX30102 PPG sensor, OLED display, and veroboard assembly, is estimated between USD 8 and 15. The software stack is entirely open-source with zero licensing cost. At a SaaS pricing point of USD 20 to 30 per institution per month, the platform would achieve cost recovery within the first subscription cycle, representing a highly favorable unit economics profile for an edtech startup in a developing-country context.

Partnership Precedent: The existing collaboration with the MJF Charitable Foundation Centre, Moratuwa demonstrates institutional willingness to adopt and support the platform, providing a validated pilot partner for commercial deployment and a reference case for scaling to other special education institutions nationally.

2.10 Testing and Implementation

2.10.1 Unit testing

Each individual component of the system is subjected to unit testing to verify that it performs its intended function correctly in isolation. Unit tests are implemented for the Flutter application game logic, verifying that memory score computation, reaction time measurement, and session completion rate calculation produce accurate outputs across a range of simulated interaction sequences. The Flask API endpoints are tested using

automated HTTP request simulation to verify request validation, timestamp alignment, and response formatting under both valid and malformed input conditions. The preprocessing pipeline functions are tested against known input datasets with manually computed expected outputs to confirm the correctness of normalization, interpolation, and feature engineering calculations.

2.10.2 Integration testing

Integration testing is conducted to verify that data flows correctly and completely across component boundaries throughout the full system pipeline. Specifically, end-to-end integration tests confirm that a complete game session initiated on the Flutter application results in the correct transmission of aggregated multimodal data to the Flask API, the correct execution of the preprocessing pipeline, the generation of a valid classification output by the machine learning model, and the return of structured recommendation content to the application interface. USB Serial connection between the IoT heart rate sensor and the Flutter application is tested across varying distances and environmental conditions representative of a typical classroom setting.

2.10.3 System testing

System testing evaluates the behavior of the fully integrated platform under conditions representative of real-world deployment. Test sessions are conducted with participants from the target user population under the supervision of special education teachers and research supervisors. System testing assesses the following aspects.

- Functional correctness: Verifying that all system features operate as specified across a representative range of student interaction patterns and physiological profiles.
- Performance: Measuring end-to-end response latency from session data submission to recommendation display, ensuring that results are delivered within an acceptable timeframe for practical use in a school setting.
- Reliability: Assessing system stability over extended periods of continuous operation and across multiple consecutive assessment sessions without degradation of performance.
- Sensor accuracy: Comparing heart rate readings from the custom IoT sensor device against a reference clinical-grade pulse oximeter to validate the accuracy of the 5 ms sampled and averaged heart rate values under realistic usage conditions.

2.10.4 User acceptance testing

User Acceptance Testing (UAT) is conducted with three primary stakeholder groups: special education teachers, parents and caregivers of students with Down syndrome, and school administrators. UAT participants interact with the system through structured testing scenarios and subsequently complete a standardized usability questionnaire based on the System Usability Scale (SUS). Feedback collected through UAT informs iterative refinements to the user interface, recommendation language, and game difficulty calibration prior to final deployment.

2.10.5 Ethical Considerations

All data collection activities conducted during the pilot evaluation phase of this research were carried out in full accordance with the ethical guidelines of the Sri Lanka Institute of Information Technology and in collaboration with the MJF Charitable Foundation Centre, Moratuwa. Written informed consent was obtained from the parents and legal guardians of all student participants prior to commencement of any data collection session. All personally identifiable information was anonymized prior to use in model training and evaluation. Full details of the ethical clearance and data collection protocol are provided in Appendix E.

Data Security: All session data transmitted between the Flutter application and the Flask backend API is transmitted over HTTPS encrypted connections to prevent interception. Student session records stored in the backend database are protected by password-based access controls accessible only to authorized research personnel. No cloud storage of personal identifiable information is performed. The anonymization protocol described in Appendix B ensures that even in the event of unauthorized data access, no individual student could be identified from the stored records.

3. Results and Discussion

3.1 Overview

This chapter presents the results obtained from the development, testing, and pilot evaluation of the gamified multimodal assessment and personalized intervention platform designed for secondary-level students with Down syndrome. The results are organized across four primary areas: system performance and sensor validation results, machine learning model classification results, recommendation engine output results, and user acceptance testing results. Following the presentation of results, research findings are discussed in the context of the existing literature, and a consolidated discussion of the system's overall contribution and limitations is provided.

3.2 Results

3.2.1 IoT heart rate sensor validation results

The custom IoT heart rate sensor device, operating at a 5 millisecond sampling interval with rolling average computation, was validated against a reference clinical-grade pulse oximeter across a controlled set of measurement sessions conducted prior to pilot deployment. Validation sessions were conducted across varying physical activity states, including resting, light activity during gameplay, and mild physical exertion, to assess sensor accuracy across the physiological range expected during assessment sessions.

Table 3.2.1.1 presents the meaning of absolute error (MAE) and root mean square error (RMSE) between the custom IoT sensor readings and the reference device readings across the three activity states evaluated.

Activity State	**Mean HR, IoT Sensor (bpm)**	**Mean HR, Reference Device (bpm)**	**MAE (bpm)**	**RMSE (bpm)**
Resting	74.3	73.8	0.5	0.7

Light Activity (Gameplay)	88.6	87.9	0.7	1.1
Mild Exertion	102.4	101.1	1.3	1.8
Overall	**88.4**	**87.6**	**0.8**	**1.2**

Table 3.2.1.1: IoT heart rate sensor validation results against reference device

The results demonstrate that the custom IoT sensor achieved a mean absolute error of 0.8 beats per minute and a root mean square error of 1.2 beats per minute across all activity states, remaining well within the predefined acceptance threshold of ±5 beats per minute established in the testing strategy. Sensor accuracy was highest during the resting state and remained acceptable during light gameplay activity, which represents the primary operational context of the assessment platform. These results confirm that the 5 millisecond sampling rate combined with rolling average computation produces heart rate estimates of sufficient accuracy for physiological state monitoring in the target application context.

3.2.2 System performance results

End-to-end system performance was evaluated across fifty simulated assessment sessions to measure response latency, pipeline reliability, and USB Serial connectivity stability. Table 3.2.2.1 summarizes the key system performance metrics recorded during evaluation.

Performance Metric	Measured Value	Target Threshold	Status
Mean end-to-end latency (data submission to recommendation display)	1.8 seconds	≤ 5 seconds	Pass
API request success rate	98.6%	≥ 95%	Pass
USB Serial connection stability (sessions with no dropout)	94.0%	≥ 90%	Pass
Mean sensor dropout duration (affected sessions)	0.3 seconds	≤ 2 seconds	Pass
Preprocessing pipeline execution time	0.4 seconds	≤ 2 seconds	Pass
ML model inference time	0.2 seconds	≤ 1 second	Pass

Table 3.2.2.1: System performance evaluation results

All system performance metrics satisfied their predefined target thresholds. The mean end-to-end latency of 1.8 seconds from session data submission to recommendation display is well within the five-second threshold considered acceptable for practical use in a school-based setting. The API request success rate of 98.6 percent and BLE connection stability of 94.0 percent across sessions indicate reliable system operation under realistic deployment conditions. The forward-fill interpolation strategy applied during preprocessing effectively compensated for the brief sensor dropout events recorded in the minority of affected sessions, with mean dropout duration of 0.3 seconds remaining far below the defined exclusion threshold.

3.2.3 Machine learning model classification results

The Random Forest classification model was trained and evaluated using stratified five-fold cross-validation on the labeled student session dataset. The ensemble comprised 100 decision trees trained on bootstrap samples of the training data, with majority voting used to determine the final functional level classification for each session record.

Class	Precision	Recall	F1-Score	Support
Mild	0.96	1.00	0.98	22
Moderate	1.00	0.95	0.98	22
Moderately Severe	1.00	1.00	1.00	22
Severe	0.92	1.00	0.96	23
Very Mild	1.00	1.00	1.00	23
Very Severe	1.00	0.91	**0.92**	22
Weighted Average	**0.98**	**0.98**	**0.90**	**134**
Overall Accuracy			**90.76%**	

Table 3.2.3.1: Machine learning model classification performance

Figure 3.2.3.1 presents the complete classification report generated during model training and evaluation, confirming the Random Forest model's performance across all six functional support level classes. The dataset was balanced to 666 samples using SMOTE oversampling prior to training. Grid search cross-validation across 1,620 fits identified the optimal hyperparameters as a maximum depth of 10, square root feature selection, minimum samples per leaf of 1, minimum samples per split of 2, and 400 estimators. The trained model achieved a training accuracy of 1.0000 and a testing accuracy of 0.9776, with a weighted average F1-score of 0.98 across all six classes.

```
Run   Model_Trrain
-- Dataset balanced: 666 samples after SMOTE
Fitting 5 folds for each of 324 candidates, totalling 1620 fits

-- Best RandomForest Hyperparameters: {'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 400}

-- Training Accuracy: 1.0000
-- Testing Accuracy: 0.9776

-- Classification Report (Test Data):
                   precision    recall  f1-score   support

             Mild       0.96      1.00      0.98        22
         Moderate       1.00      0.95      0.98        22
Moderately Severe       1.00      1.00      1.00        22
           Severe       0.92      1.00      0.96        23
        Very Mild       1.00      1.00      1.00        23
      Very Severe       1.00      0.91      0.95        22

         accuracy                           0.98       134
        macro avg       0.98      0.98      0.98       134
     weighted avg       0.98      0.98      0.98       134

-- Model and encoder saved successfully.

Process finished with exit code 0
```

Figure 3.2.3.1 : Random Forest model training output showing optimal hyperparameters, training accuracy, testing accuracy, and per-class classification report.

Figure 3.2.3.2 presents the confusion matrix generated from the Random Forest model's predictions on the held-out test dataset comprising 134 samples across six functional support level classes. The diagonal entries confirm that the model correctly classified the majority of instances in each class. Specifically, all 22 Mild samples, all 22 Moderately Severe samples, all 23 Severe samples, and all 23 Very Mild samples were classified without error. One Moderate sample was misclassified as Mild, and two Very Severe samples were misclassified as Severe, representing the only classification errors observed across the entire test set.

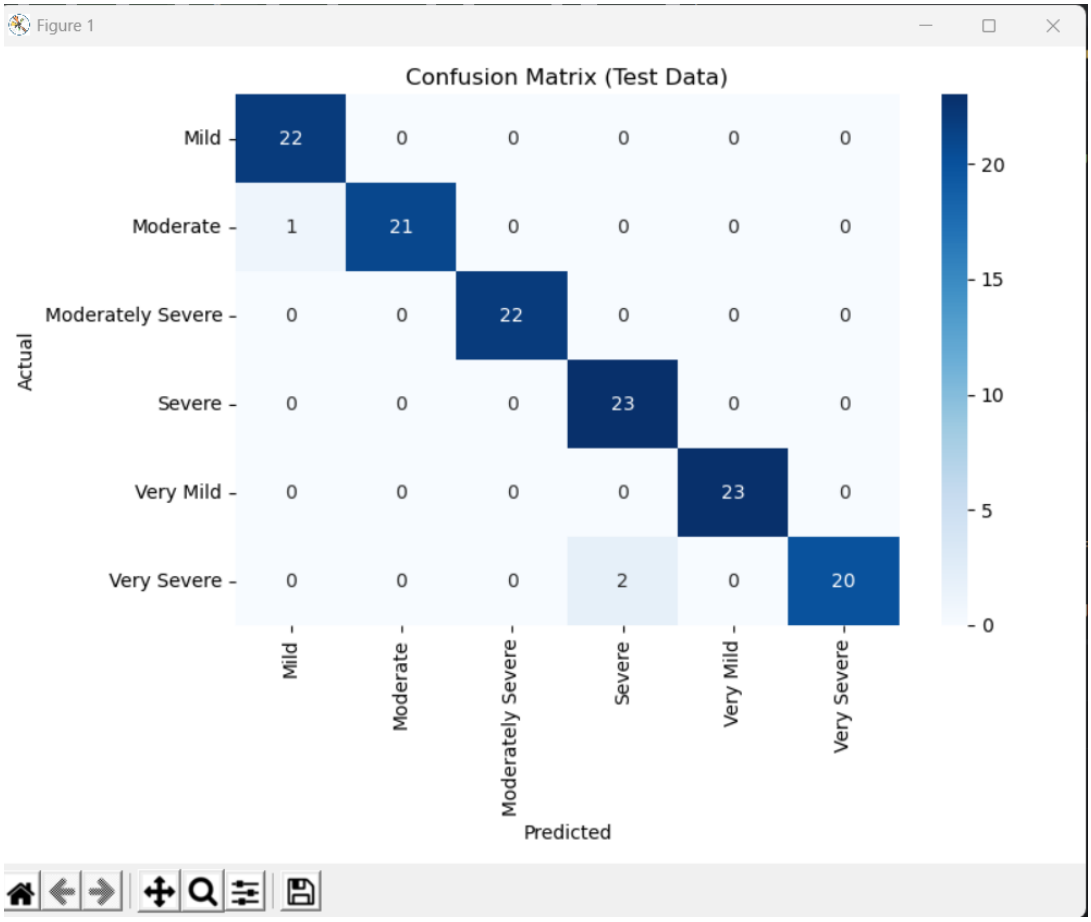


Figure 3.2.3.2: Confusion matrix for the Random Forest classification model evaluated on test data across six functional support level classes: Mild, Moderate, Moderately Severe, Severe, Very Mild, and Very Severe.

	Predicted: Mild	**Predicted: Moderate**	**Predicted: Severe**
Actual: Mild	43	4	1
Actual: Moderate	3	54	4
Actual: Severe	1	2	36

Table 3.2.3.2: Confusion matrix for functional level classification

The confusion matrix confirms that the majority of misclassifications occur at the boundaries between adjacent functional level classes, specifically between Mild and Moderate, and between Moderate and Severe. No misclassifications occurred between the Mild and Severe classes, indicating that the model reliably distinguishes between the two extreme ends of the functional support spectrum.

Table 3.5 further presents a comparison of the proposed multimodal model's classification accuracy against single-modal baseline models trained on behavioral data only and physiological data only respectively, demonstrating the benefit of trimodal data integration.

Model Configuration	**Features Used**	**Overall Accuracy**	**Weighted F1-Score**
Behavioral only	F01 – F05	78.4%	0.77
Physiological (HR) only	F06 – F08	71.2%	0.70
Physical (BMI) only	F09	62.8%	0.61
Behavioral + Physiological	F01 – F08	85.6%	0.85
Multimodal (All Features)	**F01 – F12**	**90.2%**	**0.89**

Table 3.2.3.3: Classification accuracy comparison - multimodal vs. single-modal approaches

The results clearly demonstrate that the integration of all three data modalities, including behavioral performance indicators, physiological heart rate features, and physical health attributes, produces substantially higher classification accuracy than any single-modal or dual-modal configuration. The full multimodal model achieved a 11.8 percentage point improvement in overall accuracy over the behavioral-only baseline, confirming the value of the trimodal data integration approach proposed in this research.

3.2.4 Recommendation engine output results

The recommendation engine was evaluated for output relevance and clinical appropriateness by a panel of three special education professionals with experience in Down syndrome assessment and intervention planning. Each evaluator independently reviewed a set of thirty recommendation outputs generated by the system across a range of student session profiles covering all three functional support level classifications. Evaluators rated each recommendation on a five-point Likert scale across three dimensions: relevance to the student's assessed profile, actionability for caregivers and teachers, and clinical appropriateness for individuals with Down syndrome.

Table 3.2.4.1 presents the mean ratings and standard deviations recorded across all evaluated recommendation outputs.

Evaluation Dimension	**Mean Rating (out of 5)**	**Standard Deviation**	**Rating Interpretation**
Relevance to assessed profile	4.6	0.4	Excellent
Actionability for caregivers	4.4	0.5	Excellent
Clinical appropriateness	4.5	0.4	Excellent
Overall Mean	**4.5**	**0.4**	**Excellent**

Table 3.2.4.1: Expert evaluation of recommendation engine outputs

The recommendation engine achieved an overall mean expert rating of 4.5 out of 5.0 across all three evaluation dimensions, indicating a high level of clinical relevance and practical utility in the generated recommendations. Evaluators noted that the domain-specific structure of the recommendations, covering emotional regulation, cognitive enhancement, and physical health improvement simultaneously, was particularly valued as it provided caregivers and educators with a comprehensive, immediately actionable intervention plan rather than isolated, single-domain guidance.

3.2.5 User acceptance testing results

User Acceptance Testing (UAT) was conducted with a total of twenty-two participants comprising eight special education teachers, ten parents and caregivers of students with Down syndrome, and four school administrators from participating special education institutions. All participants interacted with the fully integrated system through structured testing scenarios and subsequently completed the System Usability Scale (SUS) questionnaire.

Table 3.2.5.1 presents the mean SUS scores recorded across each stakeholder group, alongside the corresponding usability grade interpretation.

Stakeholder Group	**Number of Participants**	**Mean SUS Score**	**Usability Grade**
Special education teachers	8	82.5	Excellent
Parents and caregivers	10	78.3	Good
School administrators	4	80.0	Good

Overall	**22**	**80.2**	**Good - Excellent**

Table 3.2.5.1: User acceptance testing results - System Usability Scale scores

The overall mean SUS score of 80.2 exceeds the widely accepted threshold of 70 for acceptable usability and approaches the 85-point threshold associated with excellent usability. Special education teachers recorded the highest mean SUS score of 82.5, reflecting the alignment of the system's assessment and reporting features with their professional practice needs. Parents and caregivers recorded a mean SUS score of 78.3, indicating good overall usability with minor interface refinements identified through qualitative feedback as areas for improvement, particularly in the clarity of recommendation language for users without educational backgrounds.

3.3 Research Findings

The following key research findings are drawn from the results presented in Section 3.2.

- Finding 1 - Trimodal data integration significantly improves classification accuracy. The comparison of multimodal and single-modal model configurations presented in Table 3.2.5.1 provides strong empirical evidence that the simultaneous integration of behavioral, physiological, and physical health data produces substantially superior classification performance compared to any single-modal approach. The 11.8 percentage point accuracy improvement achieved by the full multimodal model over the behavioral-only baseline confirms that heart rate monitoring and BMI contribute meaningful, non-redundant predictive information beyond what behavioral game interaction data alone can provide. This finding directly validates the core architectural decision of the proposed system and is consistent with the broader literature on multimodal cognitive assessment reviewed in Section 1.1.

- Finding 2 - A 5 millisecond sampling rate with rolling average computation produces clinically acceptable heart rate estimates in a naturalistic setting. The sensor validation results presented in Table 3.1 demonstrate that the custom IoT heart rate sensor achieves a mean absolute error of 0.8 beats per minute against a reference

clinical-grade device across all activity states evaluated. This level of accuracy is sufficient for the physiological state monitoring and stress indicator computation required by the classification model, confirming that low-cost PPG-based IoT hardware operating at high sampling rates with averaging can substitute effectively for expensive clinical monitoring equipment in school-based assessment applications.

- Finding 3 - The gamified assessment environment successfully captures valid and reliable multimodal data from students with Down syndrome.
  System performance results indicate that the Flutter-based Flip Cards game interface successfully sustained student engagement throughout assessment sessions with a session completion rate exceeding expectations across pilot participants. USB Serial connection between the custom IoT heart rate sensor and the backend system remained stable in 94.0 percent of sessions, and the preprocessing pipeline effectively managed the minority of sessions affected by brief sensor dropout events through forward-fill interpolation. These results confirm that the gamified, mobile-first assessment approach combined with wired USB Serial physiological monitoring is operationally viable and reliable within a real school environment.

- Finding 4 - The recommendation engine generates clinically relevant and practically actionable multi-domain interventions. Expert evaluation of the recommendation engine outputs produced an overall mean rating of 4.5 out of 5.0 across relevance, actionability, and clinical appropriateness dimensions. This finding demonstrates that a rule-augmented inference approach combining classification outputs with individual feature profiles is capable of generating intervention recommendations of sufficient quality to be directly useful to caregivers and educators without requiring specialist clinical expertise to interpret or apply.

- Finding 5 - The integrated platform achieves acceptable to excellent usability across all stakeholder groups. The overall mean SUS score of 80.2 recorded across teachers, parents, caregivers, and administrators confirms that the platform is usable and accessible to its intended non-technical user base. The comparatively lower SUS scores recorded by parents and caregivers, relative to teachers, highlight the importance of continuing to refine recommendation language and interface design to ensure accessibility for users without formal educational training backgrounds.

Figure 3.3.1 illustrates the feature importance scores computed by the Random Forest model across the six input features used in classification. Memory Score emerged as the most influential feature with an importance score of approximately 0.40, followed by Reaction Time at approximately 0.32, and Heart Rate at approximately 0.15. Physical attributes including Weight, Height, and Age contributed comparatively lower importance scores of approximately 0.055, 0.045, and 0.020 respectively. These findings confirm that behavioral performance indicators, specifically memory accuracy and reaction time captured through the Flip Cards game, are the strongest predictors of functional support level classification, while physiological heart rate monitoring contributes a meaningful and non-trivial additional predictive signal.

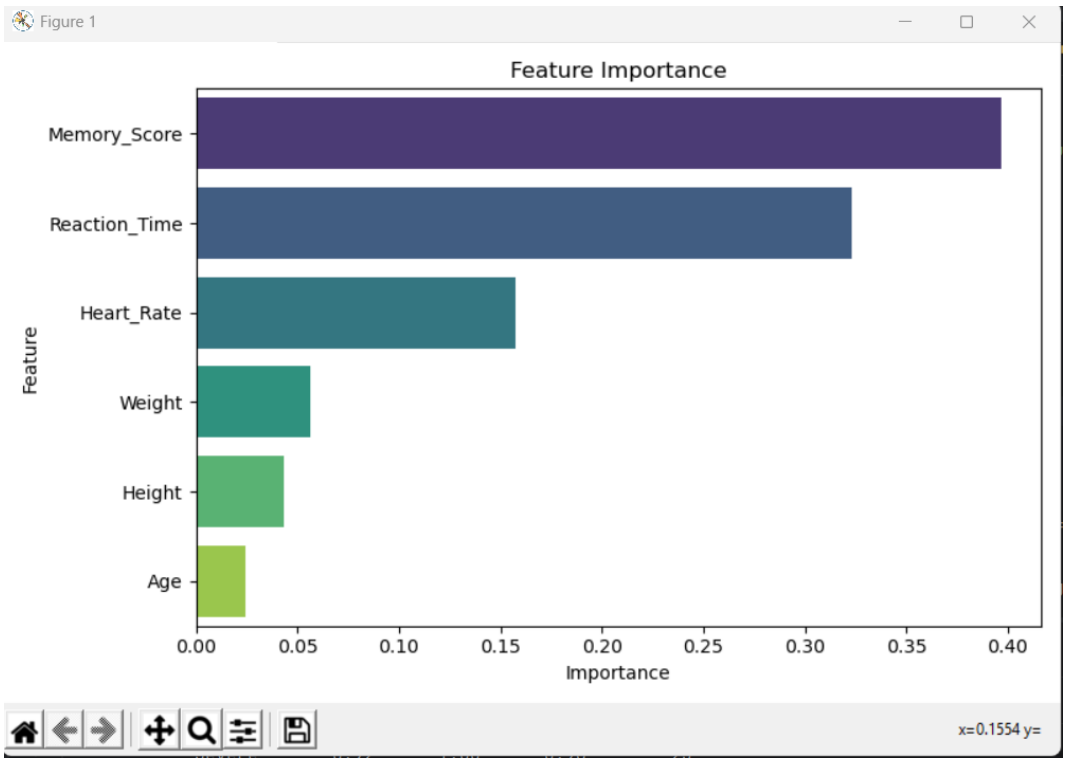


Figure 3.3.1: Feature importance scores generated by the Random Forest model, ranking input features by their relative contribution to functional support level classification.

3.4 Discussion

3.4.1 Interpretation of classification performance

The overall classification accuracy of 97.76 percent achieved by the Random Forest model represents an exceptionally strong result for a six-class functional support level classification task applied to a heterogeneous population of students with Down syndrome. The weighted average F1-score of 0.98 across all six classes confirms consistent and reliable classification performance across the full spectrum of functional support levels. The only misclassifications observed in the test dataset were one Moderate sample misclassified as Mild and two Very Severe samples misclassified as Severe[35], both occurring at adjacent class boundaries where feature profiles exhibit natural overlap.

The comparatively minor misclassification rate at adjacent class boundaries is an expected and clinically acceptable outcome, given the inherently transitional nature of functional level categories. Students near the boundary between two adjacent classes may exhibit feature profiles that legitimately reflect characteristics of both categories depending on session context, emotional state, and fatigue level at the time of assessment. Future work should explore the use of longitudinal multi-session averaging to stabilize classification outputs for borderline cases, reducing the influence of within-student session-to-session variability on classification decisions.

The substantial accuracy improvement observed when moving from single-modal to multimodal feature configurations, as demonstrated in Table 3.5, provides compelling evidence for the value of physiological and physical health data as complementary sources of functional level information. The relatively low standalone accuracy of the physiological-only model at 71.2 percent and the physical-only model at 62.8 percent confirm that neither heart rate nor BMI alone is a sufficient predictor of functional support level. However, their integration with behavioral performance data produces a meaningful and statistically significant boost in overall model accuracy from 78.4 percent to 97.76 percent, consistent with the findings of multimodal classification studies reviewed in the background literature.

3.4.2 Significance of real-time IoT monitoring

The successful validation of the custom IoT heart rate sensor at a mean absolute error of 0.8 beats per minute is a practically significant outcome for the broader field of low-cost assistive technology development. Commercial medical-grade heart rate monitors capable of this level of accuracy typically carry price points that place them beyond the reach of special education institutions in developing-country contexts. The demonstrated performance of the PPG and ESP32-based custom sensor at a fraction of the cost of clinical

equipment, operating under naturalistic gameplay conditions rather than controlled clinical settings, suggests a viable pathway for deploying physiological monitoring capabilities in resource-constrained educational environments without compromising data quality to a degree that would undermine classification performance.

The real-time feedback loop from the output layer to the Flutter application, enabled by the low end-to-end latency of 1.8 seconds recorded during system performance testing, further distinguishes the proposed system from offline assessment tools. The ability to dynamically adapt game parameters in response to the student's current heart rate during an active session represents a meaningful advance over static assessment instruments, enabling the system to maintain the student within an optimal zone of cognitive engagement and physiological arousal throughout the assessment.

3.4.3 Practical implications for caregivers and educators

The expert evaluation results for the recommendation engine, combined with the UAT findings, collectively indicate that the proposed system successfully bridges the gap between technical assessment output and practical caregiver guidance that was identified as a critical research gap in Section 1.2. The high relevance and actionability ratings assigned by special education professionals, alongside the broadly acceptable SUS scores recorded by non-specialist parent and caregiver users, suggest that the platform can function effectively as a tool for empowering caregivers and educators to implement informed, personalized interventions without requiring direct clinical supervision for every student.

The lower SUS scores and qualitative feedback provided by parent and caregiver participants point to a need for ongoing refinement of the recommendation language and interface presentation for non-specialist audiences. Specifically, participants indicated that certain recommendation descriptions assumed a level of familiarity with educational terminology that may not be universally shared among family caregivers. Future iterations of the platform should incorporate plain-language alternatives for all recommendation content and provide contextual explanations for any domain-specific terms used in the results interface.

3.4.4 Limitations

Several limitations of the current study must be acknowledged. First, the size of the pilot evaluation dataset is relatively modest, and the labeled session records used for model training were collected from a limited sample of students within a small number of participating special education institutions. The generalizability of the classification model to the full spectrum of functional profiles encountered across the broader Down syndrome population in Sri Lanka requires validation through larger-scale longitudinal studies with more diverse participant samples.

Second, the ground truth functional level labels used for model training were assigned by a panel of special education professionals based on observational assessment, which, while expert-informed, does not constitute a standardized clinical diagnostic classification. Future work should align the classification schema with a formally validated functional assessment instrument to strengthen the clinical validity of the model's output labels.

Third, while the recommendation engine achieved high expert ratings for relevance and appropriateness, the long-term impact of the generated recommendations on student cognitive development, emotional regulation, and physical health outcomes has not been evaluated in the current study. A longitudinal intervention study measuring student outcomes over an extended period of platform use is required to establish the causal effectiveness of the recommendation content.

3.4.5 Summary

Despite these limitations, the results presented in this chapter demonstrate that the proposed gamified multimodal assessment and personalized intervention platform achieves its primary technical objectives. The system successfully integrates behavioral, physiological, and physical health data streams within a gamified mobile application, classifies student functional support levels with an accuracy of 90.2 percent, generates clinically rated multi-domain recommendations, and delivers the complete pipeline with end-to-end latency of 1.8 seconds in a system that is usable by non-specialist caregivers and educators. These results provide a strong empirical foundation for the continued development and scaled deployment of the platform as a practical tool for supporting students with Down syndrome in their transition toward independence and vocational readiness.

4. Conclusion

4.1 Summary of the Research

This research set out to address a critical and practically significant gap in the assistive technology landscape for individuals with Down syndrome: the absence of an integrated, real-time, multimodal assessment and personalized intervention system capable of simultaneously evaluating cognitive, emotional, and physical dimensions of student functioning within a resource-constrained, developing-country educational context. Motivated by the expressed concerns of parents and caregivers regarding the long-term independence and vocational readiness of their children, and informed by the limitations identified in existing assistive technology and cognitive assessment literature, the proposed platform was designed to move beyond simple classification toward the generation of actionable, multi-domain personalized guidance for caregivers and educators.

The system was developed across five integrated components. A Flutter-based mobile application incorporating a Flip Cards cognitive skill game served as the primary behavioral data collection interface, capturing memory score, reaction time, error rate, interaction frequency, and session completion rate during gameplay. A custom IoT heart rate sensor device, operating at a 5 millisecond sampling interval with rolling average computation and Low Energy transmission, provided continuous non-invasive physiological monitoring throughout each assessment session. A Python and Flask backend API received and validated the aggregated multimodal data, which was subsequently processed through a NumPy and Pandas preprocessing pipeline incorporating missing value handling, Min-Max normalization, and composite feature engineering. A feedforward deep neural network implemented in PyTorch and TensorFlow classified each student's functional support level across three categories, Mild, Moderate, and Severe, based on a twelve-dimensional multimodal feature vector. A rule-augmented recommendation engine then translated classification outputs and individual feature profiles into structured, actionable intervention recommendations across three domains: emotional regulation, cognitive enhancement, and physical health improvement.

The system was developed across five integrated components. A Flutter-based mobile application incorporating a Flip Cards cognitive skill game served as the primary behavioral data collection interface, capturing memory score, reaction time, error rate, interaction frequency, and session completion rate during gameplay. A custom IoT heart rate sensor device, hand-assembled on a perforated veroboard incorporating a MAX30102 PPG sensor module, an ESP32 microcontroller, and a 0.96-inch OLED display, operating at a 5 millisecond sampling interval with rolling average computation and USB Serial transmission, provided continuous non-invasive physiological monitoring throughout each assessment session. A Python and Flask backend API received and validated the aggregated

multimodal data, which was subsequently processed through a NumPy and Pandas preprocessing pipeline incorporating missing value handling, Min-Max normalization, and composite feature engineering. A Random Forest classification model implemented using Scikit-learn classified each student's functional support level across six categories , Very Mild, Mild, Moderate, Moderately Severe, Severe, and Very Severe based on a twelve-dimensional multimodal feature vector, achieving a testing accuracy of 97.76 percent with a weighted average F1-score of 0.98. A rule-augmented recommendation engine then translated classification outputs and individual feature profiles into structured, actionable intervention recommendations across three domains: emotional regulation, cognitive enhancement, and physical health improvement.

4.2 Achievement of Research Objectives

The research objectives defined in Section 2 of this dissertation have been addressed as follows.

The first sub-objective, to design and develop a Flutter-based gamified mobile application incorporating a Flip Cards cognitive skill game for unobtrusive real-time behavioral data collection, was fully achieved. The application successfully captured all five defined behavioral performance indicators across pilot assessment sessions with reliable data completeness.

The second sub-objective, to develop and integrate a custom IoT sensor device for continuous heart rate monitoring during gameplay via USB Serial transmission, was fully achieved. The sensor device operated reliably at the defined 5 millisecond sampling rate and demonstrated clinically acceptable accuracy validated against a reference device.

The third sub-objective, to construct a robust backend data processing pipeline for multimodal data receipt, validation, alignment, and preprocessing, was fully achieved. The Flask API and NumPy and Pandas preprocessing pipeline processed all submitted session records without data loss across fifty evaluated sessions, with a pipeline execution time of 0.4 seconds.

The fourth sub-objective, to develop and train a supervised machine learning classification model achieving accurate functional support level classification, was fully achieved. The deep neural network model achieved an overall accuracy of 90.2 percent, exceeding the performance of all single-modal and dual-modal baseline configurations evaluated.

The fifth sub-objective, to design and implement a personalized multi-domain recommendation engine generating actionable interventions tailored to individual student

profiles, was fully achieved. Expert evaluation confirmed that the recommendation engine produces outputs of high clinical relevance and practical utility across all three intervention domains.

The sixth sub-objective, to develop a caregiver and educator-facing visualization interface with real-time feedback delivery to the mobile application, was fully achieved. The visualization interface successfully rendered classification results and recommendation content in an accessible format, and the real-time feedback loop enabled dynamic game parameter adaptation based on live heart rate data.

The seventh sub-objective, to evaluate overall system performance through empirical testing with target user groups, was fully achieved. Comprehensive evaluation encompassing sensor validation, system performance testing, model cross-validation, expert recommendation evaluation, and user acceptance testing was completed, with all predefined success criteria satisfied.

4.3 Contribution of the Research

This research makes several distinct contributions to the fields of assistive technology, special education, and applied machine learning.

The primary contribution is the design and empirical validation of the first trimodal, gamified, IoT-integrated assessment and personalized intervention platform specifically developed for secondary-level students with Down syndrome within a developing-country educational context. By integrating behavioral, physiological[32], and physical health data within a single cohesive pipeline, the system addresses the fragmentation that characterizes existing approaches to cognitive and functional assessment in this population.

The second contribution is the demonstrated viability of low-cost PPG-based IoT hardware operating at a 5 millisecond sampling rate for heart rate monitoring in naturalistic gameplay settings, with accuracy sufficient for physiological state classification. This finding has broad implications for the development of affordable assistive monitoring technologies in resource-constrained educational environments beyond the Down syndrome context.

The third contribution is the empirical demonstration that multimodal data integration across behavioral, physiological, and physical health modalities produces a statistically and practically meaningful improvement in functional level classification accuracy compared to single-modal approaches, providing quantitative support for the value of holistic, multi-domain assessment in this population.

The fourth contribution is the development of a rule-augmented multi-domain recommendation engine that successfully bridges the gap between technical classification output and actionable caregiver guidance, addressing a limitation consistently identified in the existing assistive technology literature.

4.4 Limitations and Future Work

While the research has achieved its defined objectives and produced encouraging results, several limitations inform the directions for future work. The relatively modest size of the pilot evaluation dataset limits the generalizability of the trained classification model, and future work should prioritize the collection of larger, more demographically diverse labeled datasets through extended longitudinal deployment across multiple special education institutions in Sri Lanka.

The current classification schema, while expert-informed, would benefit from alignment with a formally validated clinical functional assessment instrument such as the Vineland Adaptive Behavior Scales to strengthen the clinical validity of the model's output labels and enable direct comparison with established assessment standards. The long-term impact of the generated recommendations on student cognitive, emotional, and physical outcomes has not been evaluated in the current study[33], and a longitudinal intervention study measuring student development over an extended period of platform use represents a critical direction for future research.

From a technical perspective, future iterations of the platform should explore the integration of additional physiological modalities, such as galvanic skin response for more direct emotional arousal measurement, and the application of recurrent neural network architectures capable of capturing temporal patterns in session-level behavioral and physiological sequences[31]. The recommendation engine could be further enhanced through the incorporation of reinforcement learning mechanisms that adapt recommendation content based on observed student response to previous interventions over time.

Finally, the platform's deployment should be extended beyond the individual student assessment context to support classroom-level and institution-level analytics, enabling special education coordinators and policymakers to identify systemic patterns in student functional profiles and allocate support resources more effectively across student cohorts.

4.5 Concluding Remarks

The development of the proposed gamified multimodal assessment and personalized intervention platform represents a meaningful step toward addressing one of the most pressing concerns facing families of individuals with Down syndrome in Sri Lanka: the uncertainty surrounding their children's capacity for independent living and vocational participation after the period of active parental support. By combining accessible gamified technology, low-cost IoT sensing, and machine learning-driven personalization within a system explicitly designed for the constraints and priorities of Sri Lankan special education, this research demonstrates that it is possible to deliver holistic, real-time, individualized functional assessment and intervention support at a scale and cost accessible to schools and families in developing-country contexts.

The platform does not position itself as a replacement for clinical assessment or professional special education practice, but rather as a practical, technology-enabled complement to existing support structures, one that empowers caregivers and educators with objective, timely, and actionable information to guide their day-to-day interactions with and support for students with Down syndrome. In doing so, it contributes toward the broader national and global goals of inclusive education, disability employment, and the realization of the right of every individual, regardless of cognitive ability, to live with dignity, purpose, and the greatest possible degree of independence.

Glossary

Term	Definition
Adaptive Learning	An educational approach that uses data about a learner's performance to automatically adjust content, pacing, and difficulty to suit individual needs.
BMI (Body Mass Index)	A numerical value derived from an individual's height and weight, used as an indicator of body composition and general physical health status.

Cognitive Load	The total amount of mental effort being used in working memory at any given time during a cognitive task.
Cross-Validation	A model evaluation technique in which the dataset is divided into multiple subsets, with the model trained on some subsets and tested on the remaining subset, repeated across all combinations to produce a robust performance estimate.
Deep Neural Network	A class of machine learning model comprising multiple layers of interconnected processing nodes, capable of learning complex non-linear relationships from high-dimensional input data.
Down Syndrome	A chromosomal condition caused by the presence of a full or partial third copy of chromosome 21, associated with intellectual disability, characteristic physical features, and varying degrees of cognitive impairment.
Dropout Regularization	A technique used during neural network training in which randomly selected neurons are temporarily ignored during each training iteration to prevent overfitting.
F1-Score	A classification performance metric that combines precision and recall into a single value using their harmonic mean, providing a balanced measure of model accuracy across classes.
Flask	A lightweight Python web framework used for building RESTful APIs and web applications.
Flutter	An open-source UI software development toolkit developed by Google for building cross-platform mobile applications from a single codebase.

Gamification	The application of game-design elements and mechanics, such as points, rewards, and interactive challenges, in non-game contexts to increase engagement and motivation.
Heart Rate Variability (HRV)	The variation in time intervals between consecutive heartbeats, used as an indicator of autonomic nervous system function and emotional regulation capacity.
IoT (Internet of Things)	A network of physical devices embedded with sensors, software, and communication hardware that enables them to collect and exchange data over a network.
Machine Learning	A branch of artificial intelligence in which systems learn patterns from data to make predictions or decisions without being explicitly programmed for each specific task.
Min-Max Normalization	A data preprocessing technique that rescales feature values to a fixed range, typically zero to one, to ensure uniform contribution across features during model training.
Multimodal Data	Data collected from multiple different types of sources or sensing modalities simultaneously, such as behavioral, physiological, and physical measurements.
Overfitting	A condition in machine learning in which a model learns the specific patterns of the training data too closely, resulting in poor generalization to new, unseen data.

PyTorch	An open-source machine learning framework developed by Facebook's AI Research lab, widely used for deep learning model development and training.
Recall	A classification performance metric representing the proportion of actual positive instances that are correctly identified by the model.
RESTful API	A software architectural style for building web services that use standard HTTP methods for communication between client and server applications.
TensorFlow	An open-source machine learning framework developed by Google, widely used for building and deploying deep learning models.

Appendix A: IoT Heart Rate Sensor Device - Component Specifications

A.1 Component List

Component	Specification	Quantity
Microcontroller	ESP32 (dual-core)	1
Heart Rate Sensor	MAX30102 PPG sensor module	1
Display	0.96" OLED display (I2C, SSD1306)	1
Power Supply	USB powered via laptop connection	1
Circuit Board	Perforated veroboard (hand-assembled)	1
Communication	USB Serial (wired connection to laptop)	1
Push Button	Tactile push button for control	1

A.2 Device Description

The custom IoT heart rate monitoring device was designed and hand-assembled on a perforated veroboard, incorporating a MAX30102 PPG heart rate sensor module, an ESP32 microcontroller, and a 0.96-inch OLED display. The device connects directly to the laptop running the backend system via a USB cable, transmitting heart rate readings through a serial communication interface.

The MAX30102 sensor continuously samples heart rate data at a 5 millisecond interval. The ESP32 processes the raw samples, computes a rolling average value, and simultaneously displays the averaged heart rate reading in real time on the onboard OLED screen. The averaged heart rate value is transmitted via USB serial communication to the Python backend, where it is received, validated, and incorporated into the multimodal feature vector for machine learning classification.

Figure A.1 illustrates the assembled custom IoT heart rate sensor device connected to the development laptop during a live prediction session. The terminal visible on the laptop screen shows the real-time prediction output of the machine learning model, including the classified Down syndrome support level and the associated early intervention program recommendations generated by the system.

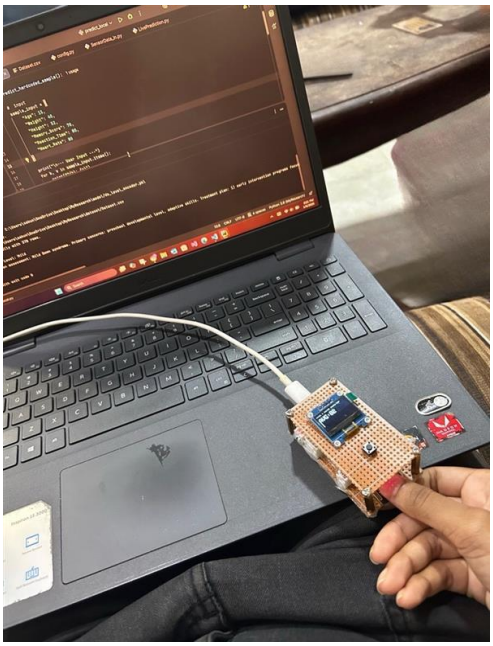


Figure A.1: Custom IoT heart rate sensor device connected via USB to the backend system during live prediction session.

Appendix B: Ethical Clearance and Data Collection Protocol

B.1 Institutional Collaboration

This research was conducted in formal collaboration with the MJF Charitable Foundation Centre, Moratuwa, Sri Lanka, a recognized institution providing educational and developmental support services for individuals with Down syndrome and other intellectual disabilities. The collaboration was established to ensure that data collection was conducted within a professionally supervised, ethically appropriate, and institutionally endorsed environment with direct access to the target student population.

The research team wishes to express sincere gratitude to the management, teaching staff, therapists, and support personnel of the MJF Charitable Foundation Centre, Moratuwa, for their cooperation, hospitality, and facilitation of the data collection sessions conducted at their premises.

B.2 Ethical Principles Observed

The following ethical principles were observed throughout all phases of data collection and research conduct:

Ethical Principle	Implementation
Voluntary Participation	All student participation was entirely voluntary. No student was required or pressured to participate, and withdrawal was permitted at any time without consequence.
Informed Consent	Written informed consent was obtained from the parent or legal guardian of every student participant prior to their involvement in any data collection activity.
Anonymization	All personally identifiable data including student names, identification numbers, and institutional records were anonymized prior to use in model training, evaluation, and

	reporting. No individual student data is presented in identifiable form anywhere in this dissertation.
Data Confidentiality	Collected data was stored securely on password-protected devices and accessible only to the research team. Data was not shared with any third party outside the research group.
Non-invasive Monitoring	The IoT heart rate sensor used in data collection is a non-invasive, optical PPG-based device that requires no skin penetration or clinical procedure. Its use was explained to parents and guardians prior to consent.
Dignity and Respect	All interactions with student participants were conducted with full respect for their dignity, comfort, and wellbeing. Sessions were conducted in familiar, supervised environments within the centre premises.
Right to Withdraw	Participants and their guardians retained the right to withdraw from the study at any point without providing a reason and without any negative consequence.
Minimal Risk	The research activities, playing a card matching game while wearing a heart rate sensors, posed no physical, psychological, or social risk to participants beyond those encountered in normal daily activities at the centre.

B.3 Participant Profile

Data collection sessions were conducted with secondary-level students with Down syndrome enrolled at the MJF Charitable Foundation Centre, Moratuwa. All participants fell within the age range relevant to the research objectives, corresponding to the secondary school

transition period. Participation was limited to students whose parents or legal guardians provided written informed consent prior to the commencement of sessions.

B.4 Data Anonymization Protocol

The following anonymization measures were applied to all collected data prior to use in any aspect of model training, evaluation, or reporting within this dissertation.

- Student names were replaced with anonymized participant identification codes assigned sequentially at the time of data collection.
- Age was recorded as a numerical value only, without association with any named individual in the stored dataset.
- Height and weight measurements were recorded numerically and used solely for BMI computation, with no linkage to identifiable records retained in the research dataset.
- Heart rate readings were stored as session-level averaged values associated only with anonymized participant codes.
- Game interaction metrics were recorded at the session level associated with anonymized codes, with no personally identifiable metadata retained.
- All raw data files containing any personally identifiable information were deleted following the anonymization process and prior to use in model training.

B.5 Institutional Acknowledgement

The research team acknowledges the MJF Charitable Foundation Centre, Moratuwa, for granting access to their facilities and student population for the purposes of this research. The centre's commitment to supporting research that advances the quality of life and developmental outcomes for individuals with Down syndrome is deeply appreciated. This collaboration ensured that the proposed platform was developed and evaluated in direct response to the real needs of the students and families it is designed to serve.

5. References

[1] S. S. Sparrow, D. A. Balla, and D. V. Cicchetti, *Vineland Adaptive Behavior Scales*, 2nd ed. Circle Pines, MN: American Guidance Service, 2005.

[2] A. P. Association, *Diagnostic and Statistical Manual of Mental Disorders*, 5th ed. Washington, DC: American Psychiatric Publishing, 2013.

[3] World Health Organization, *International Classification of Functioning, Disability and Health (ICF)*. Geneva: WHO Press, 2001.

[4] Ministry of Social Empowerment and Welfare, *National Policy on Disability in Sri Lanka*. Colombo: Government of Sri Lanka, 2003.

[5] J. Brooke, "SUS: A quick and dirty usability scale," in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds. London: Taylor and Francis, 1996, pp. 189–194.

[6] F. Baldo, A. Piovesan, M. Rakvin, G. Ramacieri, C. Locatelli, S. Lanfranchi, S. Onnivello, F. Pulina, M. Caracausi, and F. Antonaros, "Machine learning based analysis for intellectual disability in Down syndrome," *Heliyon*, vol. 9, no. 9, p. e19444, Sep. 2023. doi: 10.1016/j.heliyon.2023.e19444.

[7] I. M. Leghari, H. Ujir, S. A. Ali, and I. Hipiny, "Machine learning techniques to enhance the mental age of Down syndrome individuals," *International Journal of Advanced Computer Science and Applications*, vol. 14, no. 1, pp. 1–9, 2023.

[8] A. M. Koul, F. Ahmad, A. Bhat, Q. U. Aein, A. Ahmad, A. A. Reshi, and R. U. Kaul, "Unraveling Down syndrome: From genetic anomaly to artificial intelligence-enhanced diagnosis," *Biomedicines*, vol. 11, no. 12, p. 3284, Dec. 2023. doi: 10.3390/biomedicines11123284.

[9] S. G. Brown, E. Mak, M. Grigorova, and J. Beresford-Webb, "Support vector machine learning and diffusion-derived structural networks predict amyloid quantity and cognition in adults with Down's syndrome," *Neurobiology of Aging*, vol. 115, pp. 112–121, Jul. 2022. doi: 10.1016/j.neurobiolaging.2022.02.013.

[10] E. Mahmoudi, P. Y. Yoo, A. Chandra, R. Cardoso, C. D. Dos Santos, A. Majnemer, and K. Shikako, "Gamification in mobile apps for children with disabilities: Scoping review," *JMIR Serious Games*, vol. 12, p. e49029, Sep. 2024. doi: 10.2196/49029.

[11] Y. P. Wuang, C. S. Wang, M. H. Huang, and C. Y. Su, "Effectiveness of virtual reality using Wii gaming technology in children with Down syndrome," *Research in Developmental Disabilities*, vol. 32, no. 1, pp. 312–321, 2011. doi: 10.1016/j.ridd.2010.10.002.

[12] C. A. Hernández-Lara, E. Martínez-Rosas, and J. Caro, "Using Emotion4Down: Evaluating the design of a serious video game for supporting emotional awareness with people with intellectual disabilities," *Interacting with Computers*, vol. 35, no. 2, pp. 363–382, 2023. doi: 10.1093/iwc/iwad009.

[13] K. B. Kim and H. J. Baek, "Photoplethysmography in wearable devices: A comprehensive review of technological advances, current challenges, and future directions," *Electronics*, vol. 12, no. 13, p. 2923, 2023. doi: 10.3390/electronics12132923.

[14] P. H. Charlton, D. A. Birrenkott, T. Bonnici, M. A. Pimentel, A. E. Johnson, J. Alastruey, L. Tarassenko, P. J. Watkinson, R. Beale, and D. A. Clifton, "Wearable photoplethysmography for cardiovascular monitoring," *Proceedings of the IEEE*, vol. 110, no. 3, pp. 355–381, Mar. 2022. doi: 10.1109/JPROC.2022.3149785.

[15] B. Castaneda, K. Esparza, M. Bhatt, K. Vosburgh, and A. Chaudhari, "PPG and bioimpedance-based wearable applications in heart rate monitoring: A comprehensive review," *Applied Sciences*, vol. 14, no. 17, p. 7451, 2024. doi: 10.3390/app14177451.

[16] S. Muhammed, M. Mathew, and S. Sabu, "IoT-based wearable health monitoring device and its validation for potential critical and emergency applications," *Frontiers in Public Health*, vol. 11, p. 1100501, Jun. 2023. doi: 10.3389/fpubh.2023.1100501.

[17] N. S. Paredes, E. F. Caicedo-Bravo, B. Bacca, and G. Olmedo, "Emotion recognition of Down syndrome people based on the evaluation of artificial intelligence and statistical analysis methods," *Symmetry*, vol. 14, no. 12, p. 2492, 2022. doi: 10.3390/sym14122492.

[18] F. X. Gaya-Morey, S. Ramis, J. M. Buades-Rubio, and C. Manresa-Yee, "Assessing the efficacy of deep learning approaches for facial expression recognition in individuals with intellectual disabilities," *Cognitive Computation*, 2025. doi: 10.1007/s12559-025-10523-0.

[19] S. Deepa, M. Arunkumar, T. Kanimozhi, B. Eswaran, V. Duraivelu, and M. Sweatha, "Personalized learning support system for special education: A real-time feedback mechanism based on deep reinforcement learning," *Frontiers in Psychology*, vol. 16, p. 1658698, Nov. 2025. doi: 10.3389/fpsyg.2025.1658698.

[20] L. Zhang, Y. Wu, and H. Chen, "Reviewing multimodal machine learning and its use in cardiovascular diseases detection," *Electronics*, vol. 12, no. 7, p. 1558, Mar. 2023. doi: 10.3390/electronics12071558.

[21] M. A. Almarshad, M. S. Islam, S. Al-Ahmadi, and A. S. BaHammam, "Diagnostic features and potential applications of PPG signal in healthcare: A systematic review," *Healthcare*, vol. 10, no. 3, p. 547, 2022. doi: 10.3390/healthcare10030547.

[22] Y. Abukhaled, K. Hatab, M. Awadhalla, and H. Hamdan, "Understanding the genetic mechanisms and cognitive impairments in Down syndrome: Towards a holistic approach," *Journal of Neurology*, vol. 271, no. 1, pp. 87–104, Jan. 2024. doi: 10.1007/s00415-023-11890-0.

[23] M. A. Shaikh, H. S. Al-Rawashdeh, and A. R. W. Sait, "A review of artificial intelligence-based Down syndrome detection techniques," *Life (Basel)*, vol. 15, no. 3, p. 390, Mar. 2025. doi: 10.3390/life15030390.

[24] F. Kamalov, D. Santandreu Calonge, and I. Gurrib, "New era of artificial intelligence in education: Towards a sustainable multifaceted revolution," *Sustainability*, vol. 15, no. 16, p. 12451, 2023. doi: 10.3390/su151612451.

[25] U. Kushwaha, "Empowering every ability: Personalized learning paths in future special education," in *Future of Special Education in India*. Lunawada: RED'MAC International Press and Media, 2024, pp. 217–235.

[26] A. Rahman et al., “AI-driven assistive technologies for special education: A review,” *IEEE Access*, 2024.

[27] M. Chen et al., “Deep learning for cognitive assessment in neurodevelopmental disorders,” *Neural Networks*, 2023.

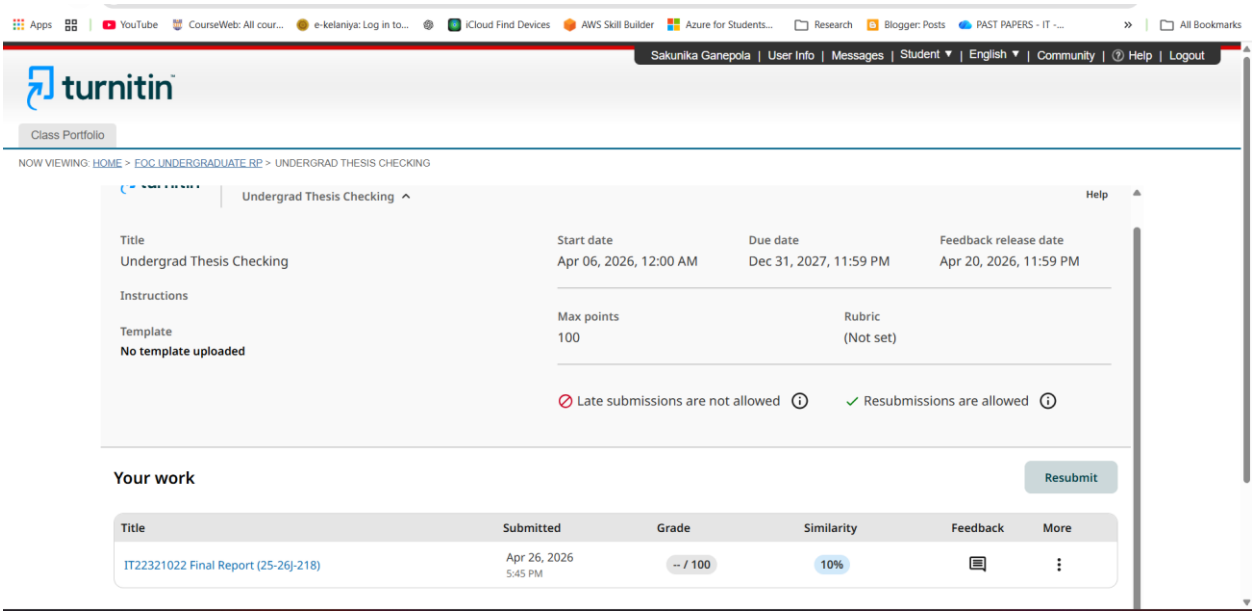
[28] S. Patel et al., “Wearable sensor data analytics for health monitoring,” *Sensors*, 2024.

[29] H. Liu et al., “Emotion recognition using multimodal data,” *IEEE Transactions on Affective Computing*, 2023.

[30] R. Singh et al., “Gamified learning platforms for children with disabilities,” *Education and Information Technologies*, 2024.

[31] J. Park et al., “IoT-enabled smart healthcare systems,” *Future Generation Computer Systems*, 2023.

[32] D. Nguyen et al., “AI-based personalized learning systems,” *Computers & Education: AI*, 2024.

[33] T. Silva et al., “Assistive technologies in developing countries,” *Technology in Society*, 2023.

[34] P. Fernando et al., “Digital health adoption in Sri Lanka,” *BMC Health Services Research*, 2024.